

Early Childhood Development and Structural Change

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Abstract

This paper presents a theory of how early childhood development shapes structural change. I document two novel cross-country facts: parental time devoted to young children (ages 0–4) and early childhood cognitive outcomes both rise with economic development, together with the High-Skill Services employment share. I construct a heterogeneous agent general equilibrium model that embeds a multi-stage human capital technology with dynamic complementarity into a structural change model with non-homothetic preferences and unbalanced sectoral TFP growth. Through dynamic complementarity, higher early childhood human capital generates a more skilled workforce, which raises both the supply of and demand for High-Skill Services. The model matches patterns of time use, structural change, and the skill distribution in the postwar US. I show that dynamic complementarity accounts for 33.1% of US structural change toward High-Skill Services in general equilibrium. The result suggests that skill formation in early childhood matters for the sectoral composition of modern economies.

JEL Codes: E24, J13, J22, J24, O11, O41

Keywords: early childhood development, structural change, high-skill services, time allocation, dynamic complementarity, skills

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1. Introduction

As countries develop, economic activity shifts away from Goods and into Services, a process known as structural change (Kuznets, 1957; Herrendorf, Rogerson and Valentinyi, 2014). Between 1975 and 2010, almost all countries in the GGDC 10-Sector Database experienced an increase in its Services employment share, and in roughly half of them the share rose by more than 20 percentage points (Timmer, de Vries and de Vries, 2015).¹ In developing countries, this shift has occurred at lower levels of income than in richer countries (Rodrik, 2016), making Services the central destination of labor for modern economies. Two theories dominate the explanation of structural change: income and relative price effects.² Less attention has gone to the supply of human capital, even though sectors differ in their skill intensity. The papers that link human capital to structural change typically model it as a single schooling decision or a cohort-level endowment (Caselli and Coleman, 2001; Herrendorf and Schoellman, 2018; Porzio, Rossi and Santangelo, 2022). In doing so, they overlook a large body of work showing that skills are formed over multiple stages of childhood (Cunha et al., 2006; Cunha and Heckman, 2007; Cunha, Heckman and Schennach, 2010).

In this paper, I address this gap by asking how early childhood development shapes the sectoral composition of aggregate output. I pay special attention to the composition of Services, distinguishing between High-Skill and Low-Skill Services. This distinction matters because narrowing cross-country income differences will require that the transition to Services be accompanied by the reallocation of labor into skill-intensive, high-productivity industries that can sustain higher wages and better living standards (Duarte and Restuccia, 2010). To answer this question, I build on existing theories of structural change by endogenizing the supply of skills through early childhood development, so that early childhood investment, human capital, income and sectoral output are jointly determined in general equilibrium.

Central to my argument is the notion of dynamic complementarity: higher levels of human capital in *earlier* stages of childhood make investments in *later* stages of childhood more productive (Cunha and Heckman, 2007). Thus, the effects of early investments on childhood human capital not only persist over the life cycle, but they also amplify the

¹These include advanced economies such as France (24.3 percentage points) and Italy (23.5), as well as developing economies such as Botswana (27.0), Brazil (26.2) and Thailand (22.8).

²The former emphasizes non-homothetic preferences that induce households to shift expenditure toward luxuries as incomes rise (Kongsamut, Rebelo and Xie, 2001; Boppart, 2014; Comin, Lashkari and Mestieri, 2021; Alder, Boppart and Müller, 2022). The latter emphasizes unbalanced sectoral TFP growth, which raises the relative price of slow-growing sectors and, given complementarities in sectoral demand, shifts activity towards them (Baumol, 1967; Ngai and Pissarides, 2007).

incentives for human capital accumulation in subsequent stages of childhood development. Higher early childhood human capital is therefore key to generating a more skilled workforce. A more skilled workforce shapes structural change by facilitating an increase in the productive capacity of the skill-intensive sector.

To motivate my theory, I document two novel cross-country empirical facts that coincide with the rise in the supply of skills (Katz and Murphy, 1992; Acemoglu and Autor, 2011) and the High-Skill Services employment share with development (Buera and Kaboski, 2012; Buera et al., 2022). First, using harmonized time use data, I show that conditional on having a young child (ages 0–4) in the household, individuals in richer countries tend to allocate more time to the childcare of young children.³ I verify that this pattern is along the intensive margin (i.e. per child) and robust across mothers and fathers. Second, I use harmonized microdata (IPUMS-MICS) to show that the cognitive outcomes of 4-year-olds improve with development.

To quantify early childhood development’s contribution to structural change, I embed this mechanism in a heterogeneous agent, dynastic overlapping generations model of growth and structural change. Parents allocate their time between work and investing in their children’s human capital, which evolves according to a multi-stage technology à la Del Boca, Flinn and Wiswall (2014). Time with children yields both a static utility benefit, which enters preferences as a luxury, and a dynamic benefit through the children’s future human capital and ultimately their utility. Labor income is spent on output from three sectors: Goods, High-Skill Services and Low-Skill Services. The sectors differ in skill intensity and exogenous TFP growth. When children become adults, their human capital is fixed and, given wages, they choose whether to work as skilled or unskilled labor.

The supply-side mechanics of the model are driven by two forces and amplified by dynamic complementarity. First, as TFP rises and incomes grow, non-homothetic preferences over consumption and time induce parents to invest more time in their children. Second, as parental human capital rises across generations, so does the productivity of each hour parents invest in their children’s human capital. Greater parental time investment improves early childhood human capital and, through dynamic complementarity, generates a more skilled workforce. This raises the productive capacity of the High-Skill Services sector. Simultaneously, the traditional demand-side drivers of structural change are at play. The sectoral consumption aggregator is non-homothetic à la Comin, Lashkari and Mestieri

³Parental time allocation is an important input for early childhood human capital. Using a structural model, Del Boca, Flinn and Wiswall (2014) find that parental time allocation from both parents is the most important input for the cognitive development of young children. The importance of parental time allocation falls with the age of the child at the expense of other inputs. Furthermore, Yum (2023) finds that both parental time allocation and other inputs are poor substitutes in early stages of childhood development. Other papers that stress the role of parental time allocation for early childhood human capital outcomes relative to other inputs include Francesconi and Heckman (2016), Del Bono et al. (2016), Bernal (2008) and Fiorini and Keane (2014).

(2021) such that as households get richer, parents bias their consumption towards luxuries and away from necessities.

In general equilibrium, the distribution of human capital is key. On the one hand, the distribution of human capital shapes the household income distribution which, together with relative prices, drives aggregate demand for High-Skill Services. On the other hand, the human capital distribution has distinct implications for the aggregate supply of High-Skill Services given its higher skill intensity. These forces jointly determine the return to skills. Together with parental preferences, this return shapes how much forward-looking parents invest in their children's human capital. Accordingly, the model captures the endogenous feedback effects between the supply- and demand-side drivers of structural change.

I then discipline the model to match patterns of structural change, time use and the skill distribution in the postwar United States (US).⁴ I first validate the model by showing that it replicates the cross-country facts: lowering initial TFP, the model's key driver of income levels, reduces both parental time in childcare and the High-Skill Services share. A 25 percent decline in initial TFP across all sectors⁵ lowers household incomes and reduces the initial High-Skill Services expenditure share by 7.4 percentage points. The gap persists across subsequent generations through both supply and demand. On the supply side, parents invest less time in their young children, both because the return to skill is lower and because, with lower incomes, they bias their time allocation less towards childcare. On the demand side, lower incomes reduce household demand for High-Skill Services. Initial TFP therefore governs the *level* of sectoral expenditure shares along the transition.

My main exercise measures the role of dynamic complementarity across stages of childhood development in shaping US structural change. It quantifies how the cross-stage amplification embedded in the human capital technology affects structural change by shaping parents' incentives to invest in their children's human capital. To isolate this, I sever the link between stages of childhood: human capital at each stage is produced from the child's initial endowment rather than from the human capital accumulated in the previous stage. Thus, investments at each stage no longer build on those made earlier. The key finding is that dynamic complementarity explains 16.0 of the 25.1 percentage point increase in the High-Skill Services share of economic activity in the postwar US in partial equilibrium (i.e. fixing relative prices and wages to their baseline values). This is equivalent to 64.0% of structural change toward High-Skill Services in the US. In general equilibrium (i.e. allowing relative prices and wages to clear markets), the contribution of dynamic complementarity

⁴In the quantitative analysis, I measure structural change using sectoral expenditure shares, which map directly into the model's household demand system. This reallocation is reflected in sectoral employment and value added shares.

⁵For context, this is broadly equivalent to the US-Zimbabwe and US-South Africa aggregate TFP gaps in the 1960s according to the Penn World Table (Feenstra, Inklaar and Timmer, 2015).

falls to 33.1% of structural change to the High-Skill Services sector. Accordingly, dynamic complementarity embedded in the human capital accumulation technology makes parental time investments in childhood quantitatively important for structural change.

An additional counterfactual exercise quantifies the role of human capital inequality in shaping structural change. The result is that a mean-preserving 35% increase in human capital dispersion parameters implies a 20.3 percentage point increase in the High-Skill Services share compared to 25.1 percentage points in the baseline over the same time horizon. A rise in inequality leads to a broad-based decline in parental time devoted to childcare of young children, driving a fall in the supply of skills in the economy.

Related Literature. This paper contributes to three strands of literature. The first strand of literature is on the sources of structural change. In this paper, I focus on the role of human capital in structural change, and therefore contribute to a growing literature on supply-side drivers of structural change (Caselli and Coleman, 2001; Porzio, Rossi and Santangelo, 2022; Herrendorf and Schoellman, 2018; Acemoglu and Guerrieri, 2008). In this respect, the closest paper to mine is Porzio, Rossi and Santangelo (2022). They model structural change in the context of an overlapping generations structure, whereby workers in different cohorts self-select into either agriculture or non-agriculture. Human capital in the model is drawn from a distribution that is a function of a cohort-specific component that evolves with the return to skills as well as an individual component that is exogenously determined. In contrast, I endogenize the human capital distribution through parental time investments, allowing my model to capture the two-way feedback between structural change and human capital. Furthermore, I enrich my framework with a multi-stage law of motion for human capital. That way, the model is well-equipped to study the effects of early childhood development on structural change.

This paper also contributes to a second strand of literature linking structural change with inequality (Comin, Danieli and Mestieri, 2020; Bárány and Siegel, 2018; Buera et al., 2022; Lee and Shin, 2017; Wiedemann, 2026). The seminal paper by Katz and Murphy (1992) argues that the rising wage premium in the US resulted from a rise in demand for skills due to skill-biased technical change. Buera et al. (2022) extend this view by relating the rise in demand for skills with structural change. I enrich these frameworks by endogenizing the feedback between the demand for skills on the one hand, and the supply of skills — assumed exogenous in these papers — on the other. Within this literature, the closest paper to mine is Wiedemann (2026) because both papers argue that inequality affects structural change. Furthermore, both papers rely on non-homothetic preferences as one explanation for the link between inequality and structural change (labeled there as the luxury/necessity channel). The two papers differ, however, in important ways. First, I focus on the role

of human capital inequality as opposed to income inequality. Second, the supply-side mechanism that shapes the link between inequality and structural change stems from dynamic complementarity embedded in the human capital technology. In [Wiedemann \(2026\)](#), the supply-side mechanism is driven by the marketization of Services which results from changes in relative prices as inequality rises.

The final strand of literature relates early childhood human capital to macroeconomic outcomes ([Yum, 2023](#); [Daruich, 2018](#); [Lee and Seshadri, 2019](#); [Manuelli and Seshadri, 2014](#); [Schoellman, 2016](#); [Caucutt and Lochner, 2020](#); [Youderian, 2019](#); [de la Croix and Licandro, 2013](#)). This literature is motivated by growing evidence that early childhood development is influential in driving life-cycle outcomes ([Deming, 2009](#); [García et al., 2017](#); [Cunha and Heckman, 2007](#); [Cunha, Heckman and Schennach, 2010](#)). Of these papers, the one closest to mine is [Schoellman \(2016\)](#), which uses variation in the ages at which Indochinese migrants arrived in the US to study the effects of an additional year of early childhood in a rich economy setting. The findings suggest that early childhood development is not a strong determinant of cross-country income differences. My paper is complementary in the sense that rather than the country environment, I focus explicitly on the importance of parental time inputs for early childhood human capital and, ultimately, structural change. This is justified by the growing body of evidence that stresses the importance of parental childcare time for early childhood human capital outcomes ([Francesconi and Heckman, 2016](#); [Del Boca, Flinn and Wiswall, 2014](#); [Del Bono et al., 2016](#); [Guryan, Hurst and Kearney, 2008](#); [Fiorini and Keane, 2014](#); [Bernal, 2008](#)).⁶

Outline. The rest of the paper is organized as follows. Section 2 presents the empirical facts that motivate the theory. Section 3 presents the full general equilibrium model with heterogeneous agents, multiple sectors and endogenous human capital accumulation. Section 4 calibrates the model while Section 5 uses the quantitative version of the model to perform the counterfactual exercises. Section 6 offers concluding remarks and policy implications.

⁶Parental time allocation is known to shape the human capital formation of children, especially in early stages of childhood. This link operates through several channels. One such channel is through language acquisition and cognitive stimulation, as evidenced by [Hart and Risley \(1995\)](#) and [Villena-Roldan and Ríos-Aguilar \(2012\)](#). Another such channel is through problem-solving abilities; through prompts and structure, parents are able to calibrate support in real time to their children such that they can solve problems beyond their unassisted capacity. See [Vygotsky \(1978\)](#) for a theoretical discussion and [Pratt et al. \(1988\)](#) and [Neitzel and Stright \(2003\)](#) for empirical evidence. A final channel operates through stress regulation. [Blair and Raver \(2012\)](#) and [Belsky and Fearon \(2002\)](#) both discuss the role of sustained sensitive caregiving and attachment security for the development of executive functions and cognitive competence of children.

2. Empirics

This section documents novel cross-country evidence on parental time investments in young children (ages 0–4) and the cognitive outcomes of 4-year-old children to motivate the general equilibrium model in Section 3.

2.1 Data Sources

This section compiles data from four sources. The first data source is the latest version (v11) of the Multinational Time Use Survey (MTUS) (Lamote de Grignon Pérez et al., 2024). The MTUS is a harmonized time use survey that contains records on how diarists spent 1440 minutes on a given single day. The MTUS has two types of data. The first is the harmonized aggregated file which contains day-level data (i.e. how much diarists spent on each activity in a single day) while the second is the harmonized episode file data which contains episode-level data (i.e. an episode is a change in activity, location, transport, co-presence or the use of technological devices). I rely on the harmonized aggregated file given the interest in aggregate time use trends. The time use data is accompanied by a number of useful variables including the age, sex and education level of the diarist as well as data on the number of household members, the number of children below the age of 18 living the household and the age of the youngest child living in the household. Accordingly, each observation in the harmonized aggregated file data corresponds to a country-year-person-day observation.

The second data source is the United Nations International Children’s Emergency Fund’s (UNICEF) Multiple Indicator Cluster Survey program accessed through the Integrated Public Use Microdata Series (IPUMS-MICS) (Bolgrien et al., 2024). IPUMS-MICS offers harmonized cross-country survey-level data on children worldwide. The dataset focuses on a diverse set of topics, including early childhood development. The IPUMS-MICS dataset is split into multiple units of analysis including births, children between the ages of 0 and 4, children between the ages of 5 and 17 and women. The early childhood development module contains data on parental time investments in children and early childhood human capital outcomes.

The third dataset is the IPUMS International (IPUMS-I) dataset (Ruggles et al., 2024). This dataset contains rich household- and person-level data that is harmonized across country-year samples. The list of variables recorded is large, ranging from nativity and birthplace to occupation and industry of work. I use the person-level census data with an explicit focus on the occupation and industry modules for information on the individual’s primary occupation and industry of work.

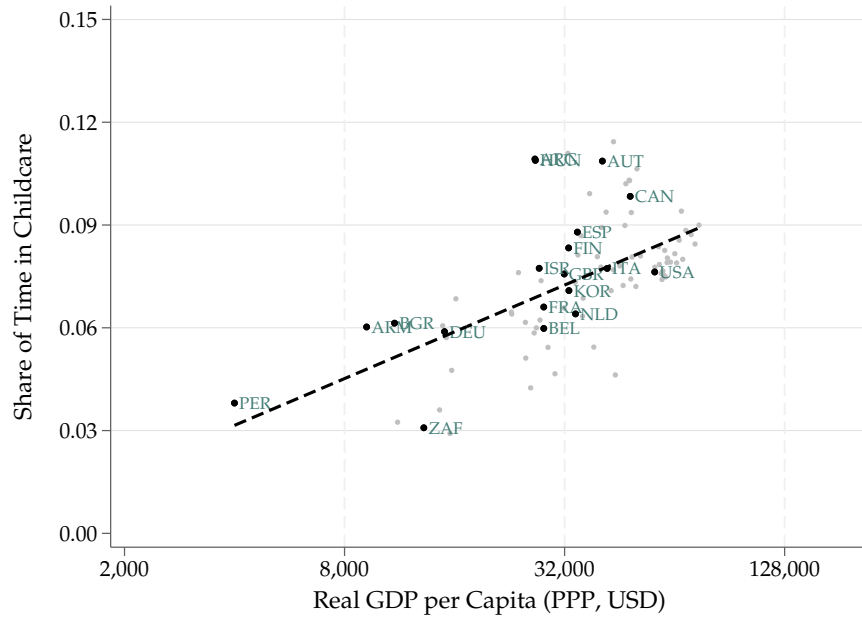
The fourth dataset is the Penn World Tables v11.0 (Feenstra, Inklaar and Timmer, 2015) which I use to obtain data on PPP-adjusted real GDP per capita at the country-year level — a standard proxy for development.

2.2 Cross-Country Patterns

To motivate my theory, I begin by documenting the first novel fact: among individuals living with a young child (ages 0–4), time spent on childcare rises with economic development. Childcare in the time use survey is defined as physical and/or medical childcare, teach and/or help with homework, read, talk and/or play with child, and supervise, accompany and/or other childcare. I restrict my attention to individuals who are between 18 and 64 years of age (i.e. working-age individuals).⁷ Figure 1 plots the country-year population-weighted average time spent on the childcare of young children by working-age individuals conditional on living with a young child against PPP-adjusted real GDP per capita. It shows that individuals in richer countries, on average, devote more time to the childcare of young children. To gauge the magnitude of these cross-country income differences, Americans earned four times as much as South Africans on average in 2010, and spent two and a half times as much time on childcare in the same year (almost two hours compared to forty two minutes).

⁷It is not possible to identify the diarist's relationship with the child (i.e. whether the diarist is the mother, father, sibling, etc. of the child). The only observables are the age of the youngest child living in the household and the number of children (below 19 years of age) in the household.

Figure 1. Share of Time Spent on Childcare



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

Figure 1 differs from other attempts to document cross-country patterns of parental time investment in childcare. Relative to [Bick, Fuchs-Schündeln and Lagakos \(2018\)](#) and [Guryan, Hurst and Kearney \(2008\)](#), I focus on early childhood human capital inputs. This means that I restrict my attention to diarists who live with at least one young child (ages 0–4) in the household rather than children under 18.

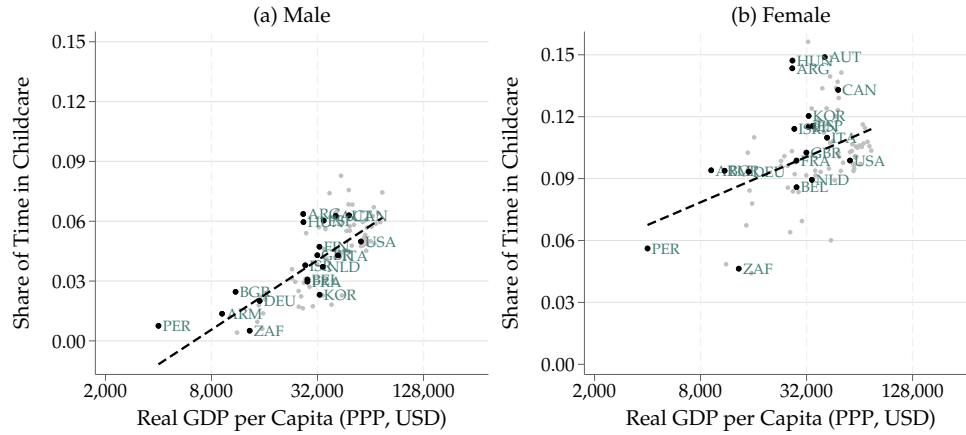
This fact can benefit from a number of robustness checks. To ensure that this trend is driven by per-child time investments, I also plot the same variable with an additional restriction that the number of children in the household is 1. Appendix A.1 shows that, qualitatively, the same trend holds, ruling out any conclusion relating this novel fact with demographic trends. Another potential challenge to Figure 1 is that while offering an expanded set of countries relative to previous attempts, it is not clear if this fact is robust across the lower end of the development spectrum. Additionally, one limitation of time use data is that it does not clarify the nature of the relationship between the individual and the child. To tackle both challenges, I present data from the IPUMS-MICS. This dataset asks mothers and fathers of children aged between 0 and 4 years of age six binary questions each on whether or not the parent in question has undertaken a given time-intensive activity with that child in the last 3 days. After aggregating the binary responses for both parents at the country-year level using children under 5 sample weights, I plot these against

PPP-adjusted real GDP per capita.⁸ The results are presented in Appendix A.2 and confirm the empirical fact that parents in richer countries dedicate more time to early childhood childcare, even on the lower end of the development spectrum. It is also helpful to confirm that these results hold within-country. While it is a well-established fact in the literature that time allocated to childcare rises over time in the US (Aguilar and Hurst, 2007; Guryan, Hurst and Kearney, 2008; Ramey and Ramey, 2010), I present the plot in Appendix A.3 using the same data as Figure 1 (focusing on children between the ages of 0 and 4 inclusive) for the sake of completeness. Barring 1975, there is a consistent increase in time spent on childcare over time implying that this phenomenon is observed both within and across countries over the development process.

Figure 2 plots the share of time spent on childcare against PPP-adjusted real GDP per capita by gender. It shows that on average both men and women dedicate more time to the childcare of young children as the economy develops. Appendix A.4 verifies that these gender-specific trends also arise from the IPUMS-MICS dataset. The difference in levels between men and women is not surprising given that these trends coincide with a number of important gender-biased structural changes accompanying the development process including changes in female labor force participation (Goldin, 1994) and home production (Bridgman, Duernecker and Herrendorf, 2018). However, that the time allocated to the childcare of young children rises between females suggests that this pattern could be separated from female labor force participation trends given that greater participation should intuitively induce a decline in time allocated to the childcare of young children. Appendix A.5 shows that the childcare time allocation share of females increase with development regardless of employment status, suggesting that this fact is robust to female labor force participation trends.

⁸The IPUMS-MICS dataset does not offer any data on minutes spent on childcare by parents, but the literature on early childhood development notes that parental time investments can be (and are frequently) measured by the frequency of activities (see Attanasio, Cattani and Meghir, 2022).

Figure 2. Share of Time Spent on Childcare By Gender



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

I then document a second novel fact: early childhood cognitive skills improve with development. To the best of my knowledge, this is the first attempt to do so at the cross-country level. To measure these outcomes, I use the IPUMS-MICS dataset which asks the child's primary caretaker three binary questions: (i) can the child read at least four simple, popular words? (ii) can the child identify at least 10 letters of the alphabet? and (iii) can the child name and recognize symbols of all numbers from 1-10? Aggregating the number of yes values at the child-level (assigning equal weights to each question) and then aggregating again at the country-year level using the proposed under-5 survey weights yields a value from 0 to 3 for each country-year observation. I plot these against PPP-adjusted real GDP per capita in Figure 3. Since the speed of cognitive development can be closely related to age when kids are youngest, I separate children that are 3 years of age and children that are 4 years of age.

Figure 3. Early Childhood Cognitive Development Demonstrated



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

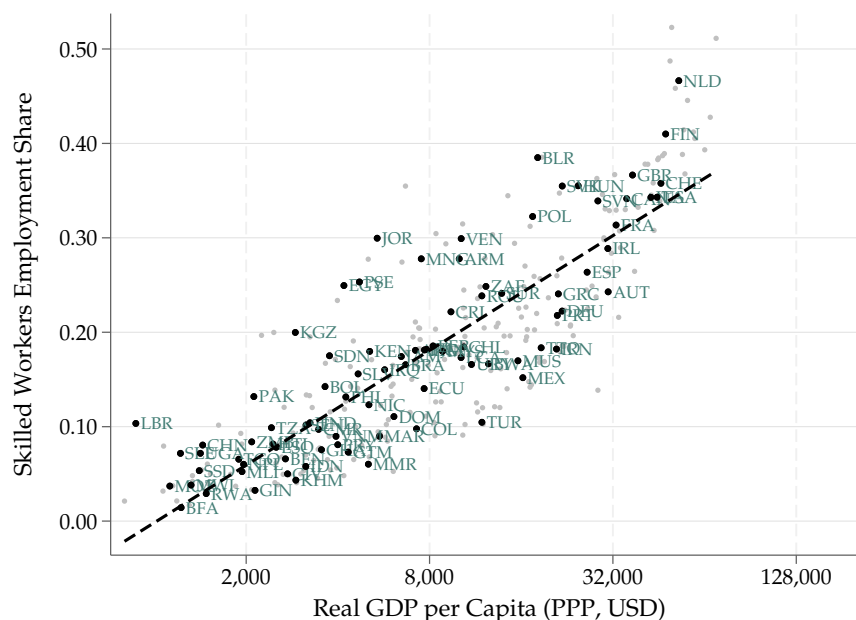
The correlation between both variables is visibly strong. In Thailand, the average PPP-adjusted real GDP per capita is roughly eight times the size of the real GDP per capita in Chad in 2019. The average Thai four-year-old answers two of the three questions correctly, while their Chadian counterpart is more likely than not to answer none. This example not only highlights that cross-country differences exist in early childhood development outcomes, but that these differences are sizable even when comparing two countries at the lower end of the development spectrum. Since the development of cognitive skills is sensitive to age when children are young, Appendix A.6 plots the same chart for three-year-old children. The results are qualitatively similar.

Thus far, I have shown that both time allocated to the childcare of young children and early childhood cognitive skills are correlated with development. To motivate the link between early childhood development and structural change, I use IPUMS-I (Ruggles et al., 2024) to show that both facts coincide with the rise in the supply of skills and the High-Skill Services employment share with development. Figure 4 plots the population-weighted share of total employment of skilled workers against PPP-adjusted real GDP per capita. Skilled workers are defined, following Acemoglu and Autor (2011), as workers who belong to managerial, professional or technical occupations.⁹ In contrast to educational

⁹These are identified as observations whose primary occupation is either legislator/senior official/manager, professional or technician/associate professional which is identified by the OCCISCO variable.

attainment — a common proxy for a worker’s skill level — the occupation captures the multi-dimensional nature of skill content, including educational attainment and wage levels. Figure 4 shows that along the development path, the share of skilled workers rises. This fact has been documented before; Katz and Murphy (1992), Acemoglu and Autor (2011) and Buera et al. (2022) are among the papers that have done so in part to understand why the wage premium rises with the relative supply of skilled labor. Figure 4 is instead helpful in showing that this phenomenon holds for a broad set of countries.

Figure 4. Skilled Workers Employment Share



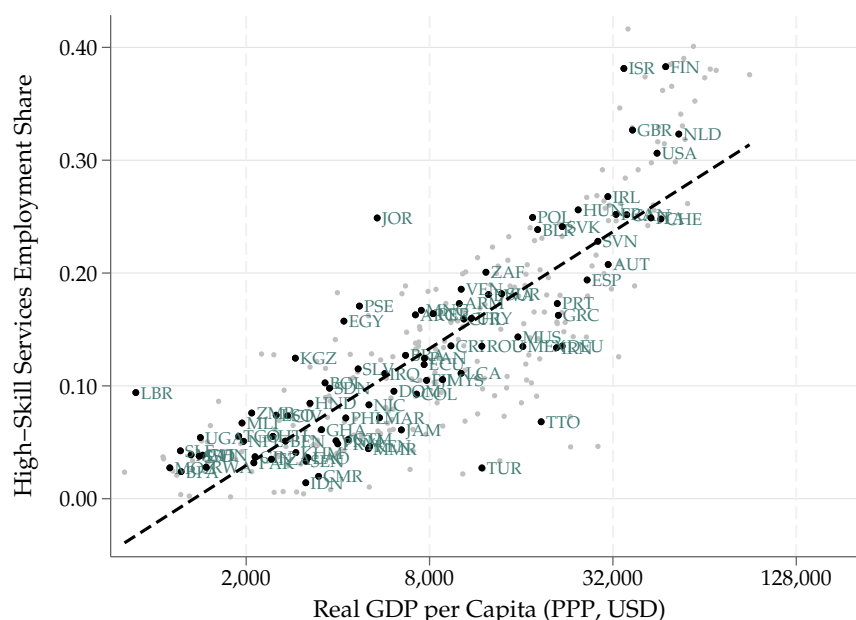
Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

I then split Services industries in the INDGEN variable into High- and Low-Skill Services sectors based on the industry’s share of total employment coming from skilled occupations, focusing only on the US censuses as the benchmark. Table A.1 in the Appendix presents the results of the industry classification by the 1960-2020 average skilled share of employment. There is a clear gap in the average skilled share of employment between the first four and the latter six Services industries, generating a natural threshold that places Education, Health and Social Work, Financial Services and Insurance and Business Services and Real Estate in the High-Skill Services bin. This classification appears consistent with every individual census year in addition to the 1960-2020 average.

Figure 5 confirms the unambiguous positive correlation between the High-Skill Services share of employment and development documented by Buera and Kaboski (2012) and Buera et al. (2022). This is not surprising given the current state of knowledge on the

process of structural transformation; the early works of Kuznets (1957), Fuchs (1968) and Baumol (1967) among others showed a shift in economic activity towards Services over the development path. More recent papers such as Buera and Kaboski (2012), Fang and Herrendorf (2021) and Buera et al. (2022) have focused exclusively on the sources and consequences of the growing importance of High-Skill Services with development, and so this is not the first paper to document the growing importance of High-Skill Services accompanying the process of growth and development. Like Figure 4 however, Figure 5 is helpful for showing that this phenomenon holds for a broad panel of countries across the development spectrum.

Figure 5. High-Skill Services Employment Share



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

2.3 Summary

This section then documents two novel empirical facts on early childhood human capital inputs and outputs. The two novel facts are that parents in richer countries on average dedicate more time to their young children and these young children perform better in cognitive tasks. This section then motivates the interest in High-Skill Services by showing that both the supply of skills and the High-Skill Services employment share rise with development, with both eventually taking up two-fifths of total employment in the richest countries.

3. Model

To uncover and quantify the causal link between early childhood development and structural change, I construct a heterogeneous agents general equilibrium framework. The modeling exercise combines an intertemporal multi-stage human capital accumulation problem into a benchmark multi-sector model of growth and structural change in order to jointly account for time use, early childhood human capital, and structural change patterns.

3.1 Environment

Assume an infinite time horizon where time is denoted by $t \in \mathbb{Z}_{\geq 0}$. At any given time t , there are three sectors that operate in this economy: Goods (G), Low-Skill Services (L) and High-Skill Services (H). Each sector has a representative firm that operates in a perfectly competitive environment, and each firm decides how many efficiency units of skilled (S) and unskilled (N) labor to hire in their respective static profit-maximization problems.

Let the economy be populated by a fixed continuum of dynasties i of measure $\mathcal{L}$ and let $g \in \{1, \dots, \infty\}$ serve as a generation superscript. At any given time t , a dynasty consists of a single parent and a single child (also referred to as a household hereafter).¹⁰ The parent is the only decision-maker inside the household. Children do not work nor consume, parents invest time to develop their child's human capital because they derive direct utility (i.e. enjoyment) from it and because they attach value to their child's future outcomes. Parental time investments are the only human capital investments allowed in this economy, and the elasticity of this input varies with the child's stage of human capital development.

Time allocated to childcare has an opportunity cost of time at work, which earns the income that parents spend on output from three sectors.¹¹ A parent working as a skilled worker contributes their human capital level as efficiency units of labor per unit of time while all parents supply a single efficiency unit of labor per unit of time if they work as an unskilled worker regardless of their human capital level and the sector they are employed in. At any given time t , parents belong strictly to one sector and occupation. Accordingly, at all times the incumbent parent of household i seeks to maximize utility by solving the following three problems: (i) a decision on how much time to invest in their child; (ii) a decision on how much to consume from each sector; and (iii) a decision on which occupation $\{S, N\}$ and sector $\{G, L, H\}$ to work in.

¹⁰This choice of demographics is motivated by the empirical facts; the rise in the share of time parents spend with their children occurs on the intensive margin (i.e. per child) and is observed for both mothers and fathers.

¹¹I therefore use the terms "worker" and "parent" interchangeably.

All households live hand-to-mouth in this economy as there are no financial markets. There is free labor mobility across both sectors and occupations for workers, and all markets clear at all times.

Timing and Heterogeneity. Each individual lives for $2T$ years. For the first T years, the individual is a child. For the subsequent T years, the individual is a parent to a child of their own.¹² The cycle repeats perpetually for each dynasty over the infinite time horizon. At any time t , an individual belonging to a given generation g is $t - [(g - 1) \times T]$ years old. Let $\mu(h, h'; t)$ denote the joint probability density function at time t of a generic parent-child human capital pair $h_i^g = h$ and $h_i^{g+1}(t) = h'$.

Upon birth, the child of dynasty i draws an initial human capital level $h_i^{g+1}(t = gT)$ from a lognormal distribution $G(\eta_c, \kappa_c)$ where η_c controls the location and κ_c controls the scale of the distribution. This distribution is time- and dynasty-invariant, and is known to all parents. Furthermore, each parent observes their child's draw upon birth. The human capital of the child from birth to age T evolves through the multi-stage law of motion for human capital accumulation, after which time the human capital level is fixed for the remainder of the individual's life. Consequently, at any time t , households are ex-ante heterogeneous in parent-child human capital pairs $h_i^g, h_i^{g+1}(t) \in \mathcal{H} \times \mathcal{H}$.

Households. At every time t , each parent with parent-child human capital pair $h_i^g, h_i^{g+1}(gT)$ is endowed with a single unit of time to be allocated between two activities: working $n_i^g(t)$ and spending time with the child $\tau_i^g(t)$. Parents seek to maximize the present discounted utility at all times. For a given dynasty, I specify the following discrete-time life-cycle structure in recursive formulation for preferences from the point of view of a parent of household i belonging to generation g

$$V_i^g(h_i^g, h_i^{g+1}(gT)) = \max_{\{\tau_i^g(t)\}_{t=gT}^{(g+1)T-1}} \left\{ \underbrace{\sum_{t=gT}^{(g+1)T-1} \beta^{t-gT} U(c_i^g(t), \tau_i^g(t))}_{\text{current generation utility}} + \underbrace{\beta^T \beta_g \mathbb{E}_{h_i^{g+2}} \left[V_i^{g+1}(h_i^{g+1}((g+1)T), h_i^{g+2}((g+1)T)) \right]}_{\text{future generations' utility}} \right\}, \quad (1)$$

¹²The life-cycle notation and structure here is similar to (Cunha and Heckman, 2007).

where β is the time discount factor, β_g is the inter-generational discount factor and $c_i^g(t)$ is the household's aggregate consumption. It is easy to see that the current formulation for preferences suggests that the parent internalizes the utility of future generations of the dynasty: the first term is the present discounted value of utility of the parent belonging to generation g while the second term is the continuation value which collects the sum of lifetime utility of *all* descendants discounted by an additional inter-generational discount factor β_g . The parent of generation g only observes the human capital draw of their child of generation $g + 1$ and not that of the generation $g + 2$ child. Accordingly, the parent takes expectations over the latter using the time-invariant initial human capital distribution $G(\eta_c, \kappa_c)$ known to the current parent.

The utility function is concave, strictly increasing and twice differentiable with $U : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. I assume the familiar [MaCurdy \(1981\)](#) structure for instantaneous utility¹³

$$U(c_i^g(t), \tau_i^g(t)) = \frac{c_i^g(t)^{1-\nu}}{1-\nu} - \omega \frac{n_i^g(t)^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}}. \quad (2)$$

The parameter ω is the time-invariant weight that the parent attaches to the pure dis-satisfaction generated from work. The choice to model a disutility for hours worked (i.e. utility from time spent with children, akin to leisure) in the first place is motivated by the findings of [Cosaert and Hennebel \(2023\)](#) who, using Dutch consumption and time use data, find that 40% of parental childcare time allocation is perceived as a form of leisure. ν is the usual CRRA parameter which controls the curvature in consumption. It can be shown that this parameter will determine the strength of income effects; if $\nu > 1$, then a rise in a given parent's income will prompt a decline in hours worked and vice versa.

Aside from their time allocation, the parent also chooses the sectoral composition of aggregate consumption. Letting $c_i^g(t)$ be a composite of consumption from three sectors: Goods ($c_{i,G}^g(t)$), Low-Skill Services ($c_{i,L}^g(t)$) and High-Skill Services ($c_{i,H}^g(t)$), I assume the following implicitly defined CES aggregator from [Comin, Lashkari and Mestieri \(2021\)](#)

$$\sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \quad (3)$$

where θ_j is a time-invariant sector-specific preference weight, $c_{i,j}^g(t)$ is the sector-specific quantity consumed and $\sigma \in (0, \infty)$ is the (constant) elasticity of substitution that

¹³This utility function choice is consistent with the wider literature on time allocation and aggregate labor supply over time (see [Bick et al. \(2022\)](#) and [Boppart, Krusell and Olsson \(2023\)](#) for some examples of this). [Boppart and Krusell \(2020\)](#) show that [MaCurdy \(1981\)](#) preferences can generate a balanced growth path with falling hours worked with income.

governs the degree of substitutability (if $\sigma > 1$) or complementarity (if $\sigma < 1$) between sectoral consumption. Crucially, λ_j influences the relative income elasticities of sectoral demand. Consequently, this will govern the extent to which the sectoral composition of the consumption basket will vary with household income. The advantage of (3) relative to alternative consumption aggregators such as Stone-Geary preferences is that, as will become clearer when studying the equilibrium path of this economy, income effects do not level off but rather persist in the long-term, better capturing long-run Engel curves.

Finally, each parent self-selects into sector $\{G, L, H\}$ and occupation of work $\{S, N\}$ at any given time t . Parents who choose a skilled occupation contribute h_i^g units of efficiency labor while those that choose an unskilled occupation contribute exactly 1 unit of efficiency labor regardless of their human capital level. Accordingly, the parent's gross income at time t , $y_i^g(t)$, belongs in the following set

$$y_i^g(t) \in \{w_{j,S}(t)h_i^g, w_{j,N}(t) : j \in \{G, L, H\}\}, \quad (4)$$

where $w_{j,S}(t)$ and $w_{j,N}(t)$ are the wage for a single unit of efficiency labor as a skilled and unskilled worker in sector j , respectively. Since households live hand-to-mouth in this economy, the following budget constraint must hold at all times

$$(1 - \tau_i^g(t)) y_i^g(t) \geq p_i^g(t) c_i^g(t), \quad (5)$$

where $p_i^g(t)$ is the price of aggregate consumption.

Human Capital Accumulation. Inspired by [Del Boca, Flinn and Wiswall \(2014\)](#) and [Cunha, Heckman and Schennach \(2010\)](#), parents have access to a multi-stage human capital production function for their children. The child's initial human capital is drawn from a time- and dynasty-invariant lognormal distribution $G(\eta_c, \kappa_c)$ where η_c and κ_c control the location and scale of the distribution, respectively. Each parent knows the distribution and observes the child's draw at all times.¹⁴ Let $s \in \{1, \dots, S\}$ denote the stage of childhood development that a child is currently in. The child's human capital evolves according to

$$h_i^{g+1}(t+1) - h_i^{g+1}(t) = h_i^{g+1}(t)^{\zeta_s + \eta} \left(h_i^g \tau_i^g(t) \right)^{1 - \zeta_s}, \quad (6)$$

where $t \in \{gT, \dots, (g+1)T - 1\}$ indexes the periods of childhood and $s = t - gT + 1 \in \{1, \dots, S\}$ denotes the corresponding stage of development. The first input $h_i^g \tau_i^g(t)$ is the

¹⁴An alternative modeling choice would be to assume that each parent does not observe the child's draw and instead makes decisions based on the expectation of the distribution such that the child's human capital is realized ex-post all investment decisions (i.e. when they become parents themselves). To remain consistent with the literature on childhood development (e.g. [Del Boca, Flinn and Wiswall, 2014](#)), I assume that the parent observes the child's draw.

parent's effective time allocation to childcare. This is composed of the units of time that the parent of household i and generation g chooses to spend with their child, $\tau_i^g(t)$, and the parent's fixed human capital stock h_i^g which acts as a productivity term to the childcare time allocation. The second input is the current human capital level of the child, $h_i^{g+1}(t)$. These inputs typically feature in multi-stage human capital models in a Cobb-Douglas fashion, as in Del Boca, Flinn and Wiswall (2014). Crucially, the relative intensities of these inputs (governed by $1 > \zeta_s > 0$) vary with the stage of childhood development, s , to capture the varying levels of importance associated with each input over the childhood development period. Relative to Del Boca, Flinn and Wiswall (2014), I add a human capital persistence term q . This regulates the degree of dynamic complementarity and therefore allows for a more flexible parameterization to fit the data.¹⁵ As soon as the child becomes a parent at the age of T , the human capital level is fixed for the remainder of life. For simplicity, there is no depreciation in human capital in this economy.

Production. On the production side¹⁶, there is a profit-maximizing sector-specific representative firm that operates in a perfectly competitive environment. The firm produces sectoral output $Y_j(t)$ by augmenting aggregate efficiency units of labor $Z_j(t)$ using a sector-specific technology $A_j(t)$ which grows at a constant growth rate g_j known to all parents at all times

$$Y_j(t) = A_j(t)Z_j(t) \quad \forall j \in \{G, L, H\}. \quad (7)$$

Sectors differ by their skill intensity: (i) Sector L hires both skilled and unskilled labor. Both types of labor are aggregated using a CES technology with an elasticity of substitution $\epsilon \in (0, \infty)$ which determines the extent to which both types of labor are substitutes (if $\epsilon > 1$) or complements (if $\epsilon < 1$) and time-invariant technology weights $0 < \phi_L < 1$ which influence the skill intensity of the sector

$$Z_L(t) = \left[\phi_L N_L(t)^{\frac{\epsilon-1}{\epsilon}} + (1 - \phi_L) S_L(t)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}}, \quad (8)$$

where $S_L(t)$ and $N_L(t)$ are the aggregate efficiency units of skilled and unskilled labor. (ii) Sector G has a technology weight of $\phi_G = 1$ and hence only hires unskilled labor

¹⁵Mathematically, dynamic complementarity requires that the cross-partial derivative of the human capital production function be positive (i.e. $\frac{\partial^2 h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t) \partial \tau_i^g(t)} > 0$). This setup implies that children with better early childhood human capital are more likely to draw in greater investments by their parents in later stages of their childhood, fixing the parent's human capital level.

¹⁶Since only the parent can work in this economy, the generation indices are dropped hereafter.

$$Z_G(t) = N_G(t). \quad (9)$$

(iii) Sector H has a technology weight of $\phi_H = 0$ and therefore only hires skilled labor

$$Z_H(t) = S_H(t). \quad (10)$$

This set-up implies that sector H is the most skilled labor-intensive sector while sector G is the most unskilled labor-intensive sector. Output from sector $j \in \{G, H, L\}$ is sold to households at a price $p_j(t)$.

3.2 Equilibrium

Sectoral Producers. The three sectoral producers choose the quantities of skilled, $S_L(t)$, and unskilled labor, $N_L(t)$, to maximize profits given their technologies, output prices and wages. The H -sector producer only hires skilled labor, and so the corresponding optimality condition reveals that the producer will hire skilled labor up until the marginal value product of labor is equated with the wage rate

$$p_H(t)A_H(t) = w_{H,S}(t). \quad (11)$$

Likewise, the G -sector producer only hires unskilled labor, and so demand for labor is also perfectly elastic with

$$p_G(t)A_G(t) = w_{G,N}(t). \quad (12)$$

The L -sector representative producer on the other hand hires both types of labor. Taking first order conditions of the L -sector producer's profit maximization problem reveals that the relative demand for skilled labor in that sector depends positively on the relative skill intensity of that sector and the relative wage of skilled labor in equilibrium

$$\frac{N_L(t)}{S_L(t)} = \left(\frac{(1 - \phi_L)}{\phi_L} \frac{w_{L,S}(t)}{w_{L,N}(t)} \right)^\epsilon. \quad (13)$$

As usual, this relationship is regulated by the elasticity of substitution between both types of labor.¹⁷

Parents. Parents decide (i) which occupation and sector to work in, (ii) how much time to invest in their child and (iii) how much of each sector to consume given preferences,

¹⁷The proof for (11), (12) and (13) is located in Appendix B.1.

wages and prices. Proposition 1 simplifies the household's problem by proving that a utility-maximizing parent is also a gross income-maximizing parent.

Proposition 1 (Separation of occupational choice). *Suppose that the occupational choice problem only affects the parent's problem through gross income. Then, the parent's problem can be solved sequentially. First, the parent makes the occupational choice that maximizes gross income. Second, the parent jointly chooses time allocation and sectoral quantities that maximize preferences (1) subject to (5) and $n_i^g(t) + \tau_i^g(t) \leq 1$.*

Proof. Any feasible allocation of time and sectoral quantity under gross income $y(t)$ remains feasible under a gross income $y'(t) > y(t)$ since $y'(t)$ expands the parent's budget without affecting preferences or the time constraint. See Appendix B.2 for formal proof. $\square$

Proposition 1 allows (4) to be written as a standard Roy selection-type problem

$$y_i^g(t) = \max \{w_{j,S}(t)h_i^g, w_{j,N}(t) : j \in \{G, L, H\}\}, \quad (14)$$

where given wages and the fixed human capital level, parents choose the sector and occupation that maximizes their gross income. With gross income determined, the parent's problem boils down to a standard two-part structural change model where $\forall t$ the given parent chooses $\tau_i^g(t)$ to maximize intertemporal preferences (1) subject to the budget constraint (5) (i.e. the intertemporal problem) and then chooses the sectoral consumption quantities that maximize (3) subject to the budget constraint (5) with net income substituted in (i.e. the static problem). The solution to the static problem yields the standard Marshallian demand functions¹⁸

$$c_{i,j}^g(t) = \frac{e_{i,j}^g(t)p_i^g(t)c_i^g(t)}{p_j(t)}, \quad \text{where} \quad e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)} \quad \forall j \in \{G, L, H\}. \quad (15)$$

The demand functions write the sectoral consumption quantity from a given sector as a function of the parent's total expenditure, $p_i^g(t)c_i^g(t)$ (which by virtue of the budget constraint is equivalent to time-adjusted income $(1 - \tau_i^g(t))y_i^g(t)$). Note that the standard income and substitution effects are present in (15). A higher relative price $p_j(t)$ reduces consumption of sector j 's good, while greater aggregate consumption induces a disproportionate change in consumption of sector j 's good by a magnitude controlled by λ_j .¹⁹

¹⁸The proof for (15) is located in Appendix B.3.

¹⁹Note that if $\lambda_j = 1$, (15) collapses to a homothetic demand function.

Equilibrium. I set the price of the H sector's output $p_H(t)$ as the numéraire in this economy.

Definition 1 (Competitive equilibrium). Normalize $p_H(t) = 1$ for all $t \in \mathbb{Z}_{\geq 0}$. A competitive equilibrium is a collection of price, wage, and technology sequences

$$\left\{ \{p_k(t)\}_{k \in \{G,L\}}, \{w_{j,o}(t)\}_{j \in \{G,L,H\}, o \in \{S,N\}}, \{A_j(t)\}_{j \in \{G,L,H\}} \right\}_{t \in \mathbb{Z}_{\geq 0}},$$

together with household allocations and firms' labor demands, such that:

1. Given prices and wages, the parent in each dynasty i and generation g , with state $(h_i^g, h_i^{g+1}(gT))$, maximizes intertemporal utility by choosing sequences of occupations, sectors of employment, sectoral consumption, and childcare time,

$$\left\{ o_i^g(t), j_i^g(t), \{c_{i,k}^g(t)\}_{k \in \{G,L,H\}}, \tau_i^g(t) \right\}_{t=gT}^{(g+1)T-1},$$

where $o_i^g(t) \in \{S, N\}$ and $j_i^g(t) \in \{G, L, H\}$, subject to the per-period budget and time constraints.

2. Given prices, wages, and technologies, the representative producer in each sector $j \in \{G, L, H\}$ maximizes profits in every period $t \in \mathbb{Z}_{\geq 0}$ by choosing its demands for skilled and unskilled labor efficiency units, $S_j(t)$ and $N_j(t)$.
3. All sectoral output markets and skilled and unskilled labor markets clear in every period $t \in \mathbb{Z}_{\geq 0}$.

3.3 Market Clearing

Structural Change. The Roy-type selection problem together with the assumption of free labor mobility across sectors for skilled and unskilled workers implies the continued relocation across sectors until sectoral wages equalize. Since both the G - and L -sector producers hire unskilled labor, unskilled labor wage equalization using these sectors' first order conditions implies

$$\underbrace{p_G(t)A_G(t)}_{w_{G,N}(t)} = \underbrace{\phi_L p_L(t)A_L(t) \left(\frac{Z_L(t)}{N_L(t)} \right)^{\frac{1}{\epsilon}}}_{w_{L,N}(t)} = w_N(t). \quad (16)$$

Wage equalization in the market for skilled labor instead uses the first order conditions from the H - and L -sector producers since they are the only sectors that hire skilled workers. The corresponding wage equalization implies

$$\underbrace{A_H(t)}_{w_{H,S}(t)} = \underbrace{(1 - \phi_L) p_L(t) A_L(t) \left(\frac{Z_L(t)}{S_L(t)} \right)^{\frac{1}{\epsilon}}}_{w_{L,S}(t)} = w_S(t). \quad (17)$$

These two equations ensure that in equilibrium and conditional on choosing an occupation, parents will be indifferent between the sector of work.

Human Capital Allocation. Likewise, the Roy-type selection problem together with the assumption of free labor mobility across occupations implies that a parent with a human capital level h_i^g will pursue skilled work if $h_i^g \geq \bar{h}(t)$ where $\bar{h}(t)$ is constructed using the indifference condition

$$\bar{h}(t) = p_G(t) \frac{A_G(t)}{A_H(t)}. \quad (18)$$

Note that the indifference condition does not depend on time allocation thanks to Proposition 1. The threshold (18) indicates that in addition to the raw human capital level of the parent, the supply of skills in the economy is also pinned down by general equilibrium effects — namely structural change (through relative prices) and sector-biased technical change (given complementarities between labor and TFP in production).

Market Clearing. Recall that $\mu(h, h'; t)$ denotes the joint probability density function at time t of a generic parent-child human capital pair $h_i^g = h$ and $h_i^{g+1}(t) = h'$ and that $\mathcal{L}$ represents the (fixed) mass of dynasties in this economy. The sectoral goods market clearing conditions can be written as

$$A_j(t) Z_j(t) = \mathcal{L} \int_h \int_{h'} c_j(h, h'; t) \mu(h, h'; t) dh' dh \quad \forall j \in \{G, L, H\}, \quad (19)$$

where the LHS is the supply and the RHS is the demand for a given sector's output. Meanwhile, the market clearing conditions for skilled and unskilled labor can be written as

$$\sum_{j \in \{G, L, H\}} S_j^d(t) = \mathcal{L} \int_{\bar{h}(t)}^{\infty} \int_{h'} h (1 - \tau(h, h'; t)) \mu(h, h'; t) dh' dh \quad \text{and} \quad (20)$$

$$\sum_{j \in \{G, L, H\}} N_j^d(t) = \mathcal{L} \int_0^{\bar{h}(t)} \int_{h'} (1 - \tau(h, h'; t)) \mu(h, h'; t) dh' dh, \quad (21)$$

where the LHS is the sum of skilled and unskilled efficiency units of labor demanded by the three firms while the RHS is the time-adjusted supply of skilled and unskilled efficiency units of labor, respectively.

3.4 Steady State

Sections 3.2 and 3.3 characterize the equilibrium conditions that must hold in the transitional dynamics of this economy. Motivated by the growing importance of High-Skill Services with development, the remainder of this section studies the properties of an economy where all activity is occupied by the H -sector.

Proposition 2 (Asymptotic Dominance of the H -sector). *Suppose that $\lambda_H > \lambda_L, \lambda_G, g_L, g_G > g_H > 0$ and that $0 < \sigma < 1$. For $k \in \{L, G\}$ define*

$$x_k \equiv -\sup_{t \geq 0} \left[(1 - \sigma)(g_H - g_k) + (1 - \sigma)(\lambda_k - \lambda_H) \frac{\dot{c}_i^g(t)}{c_i^g(t)} \right] > 0.$$

- *If $\epsilon = \infty$: the parameter restrictions are sufficient for asymptotic dominance of the H -sector.*
- *If $\epsilon < \infty$: the H -sector attains asymptotic dominance provided that, in addition, for each $k \in \{L, G\}$,*

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t \frac{(1 - \sigma)}{\epsilon} \left(\frac{\dot{S}_L(s)}{S_L(s)} - \frac{\dot{X}_L^k(s)}{X_L^k(s)} \right) ds < x_k,$$

where $X_L^L \equiv Z_L$ and $X_L^G \equiv N_L$.

Proof. See Appendix B.4 for the formal proof. □

Proposition 2 establishes the conditions under which as $t \rightarrow \infty$, $e_{i,H}^g(t) \rightarrow 1$ and $e_{i,L}^g(t), e_{i,G}^g(t) \rightarrow 0$. This proposition is used to close the model; the implied limiting case of the proposition is that the H -sector is the only active sector in the economy. Accordingly, this economy asymptotically hires only skilled workers and economy-wide TFP now grows at the constant growth rate g_H . Let $\tilde{x}(t)$ denote variable x in a steady state period $t \geq \mathcal{T}$. The representative firm's profit-maximizing equilibrium condition is

$$\tilde{w}_H(t) = \tilde{p}_H(t) \tilde{A}_H(t). \quad (22)$$

With $\tilde{p}_H(t)$ as the numéraire, this pins down the equilibrium wage for skilled workers

$$\tilde{w}_H(t) = \tilde{A}_H(t). \quad (23)$$

To make the model stationary in steady state, I proceed in two steps. First, I change the timing index t to s to denote the stage of childhood development. This step acknowledges

that while the generation problem is stationary, the behavior of parents — and therefore aggregate and equilibrium objects — may vary over the stages of childhood development. Second, I normalize steady state consumption and output by $\tilde{A}_H(s)$ to obtain $\tilde{c}_{i,H}^{g*}(s)$ and $\tilde{Y}_H^*(s)$, respectively. The stationarized version of the labor market clearing condition can be written as

$$\tilde{S}_H^d(s) = \mathcal{L} \int_h \int_{h'} h (1 - \tilde{\tau}(h, h'; s)) \tilde{\mu}(h, h'; s) dh' dh, \quad (24)$$

where the LHS is the demand for skilled labor and the RHS is the time-adjusted aggregate supply of skills in the economy in steady state stage s of childhood development. The corresponding sectoral good market clearing condition can be written as

$$\tilde{S}_H^s(s) = \mathcal{L} \int_h \int_{h'} \tilde{c}_H^*(h, h'; s) \tilde{\mu}(h, h'; s) dh' dh, \quad (25)$$

where the LHS is the normalized supply of H -sector output (i.e. $\frac{\tilde{Y}_H(s)}{\tilde{A}_H(s)}$) and the RHS is the normalized aggregate demand for H -sector output.

4. Mapping the Model to Data

This section discusses how the model is taken to the data. It starts by briefly discussing some assumptions underlying the numerical solution, and the remainder of this section lays out the calibration strategy and model fit. The parameters are disciplined using a combination of external calibration, direct measurement and internal calibration.

4.1 Numerical Solution

A detailed explanation of the solution algorithm is relegated to Appendix C.1. Instead, I elaborate here on two key assumptions that are required to proceed with the numerical solution. The first concerns the initial parent human capital distribution. This characterizes the model's initial conditions and is required to shoot forwards for the joint distribution. I assume that, like the initial child human capital distribution, the initial parent human capital distribution is governed by a lognormal distribution $G(\eta, \kappa)$ where η and κ are the location and shape parameters, respectively.

The second key assumption is around timing. I assume that there are three stages of childhood development (i.e. $S = 3$) and that each stage of childhood development lasts for a single time period in the model such that ($S = T = 3$). As per the model's demographic structure, this means that each individual from a given generation lives $2T = 6$ periods in total, 3 as a child and 3 as a parent. This assumption is invoked without loss of generality and purely for computational tractability. This in turn implies that

the human capital law of motion is governed by a total of four parameters including 1 human capital persistence parameter q and 3 parental time allocation elasticity parameters $\{\zeta_1, \zeta_2, \zeta_3\}$ where the subscripts denote the stage of childhood development. Empirically, I handle this by assuming that a single time period is equivalent to 5 years in the data.

4.2 External Calibration

The following 12 parameters are externally calibrated $\{\beta, \beta_g, \sigma, \chi, \epsilon, \phi_L, A_G(0), A_H(0), A_L(0), g_G, g_H, g_L\}$. Table 1 presents the values and sources of the externally calibrated parameters. The period discount factor β is assigned a value of 0.904. As in Barro and Sala-i-Martin (2004), this assumes an annual discount factor of 0.98. Since a period in the model is 5 years, the discount factor is retrieved by 0.98^5 . I use an inter-generational discount factor β_g of 0.5 which is assumed by Boar and Lashkari (2021). Together, these values suggest that parents hardly attach value to the continuation value of their grandchildren. The elasticity of substitution of 0.1 suggests strong complementarities in consumption between the different sectoral goods which is a common finding in the structural change literature (Herrendorf, Rogerson and Valentinyi, 2013; Buera and Kaboski, 2009). I use the Frisch elasticity value of 0.5 from Boppart, Krusell and Olsson (2024) to inform the value of χ .

The elasticity of substitution between skilled and unskilled workers in the L -sector production function ϵ is assigned a value of 2.9 implying gross substitutability between both types of labor. This value is obtained by Acemoglu and Autor (2011) who fit the method by Katz and Murphy (1992) to longer US time series data. The unskilled worker labor intensity of the L -sector denoted by ϕ_L is assigned a value of 0.75 to suggest that this sector uses unskilled labor more intensively than skilled labor. This closely matches the unskilled labor intensity in the US Low-Skill Services sector found in Fang and Herrendorf (2021).

Sectoral TFP in the model grows exogenously according to an exponential growth function. All sectoral TFP levels are normalized to 1.0 in the initial period. The growth rates of sectoral TFP are obtained from the 44-industry version of the Elbadri and Vizcaino (2025) dataset which runs from 1947 to 2023. Broad sector TFP growth rates are calculated from the industry-level growth rates and aggregated using Törnqvist weights. The industry classification into broad sectors is consistent with Appendix A.7. The growth rates in Table 1 are per period (i.e. per 5 calendar years) and, unsurprisingly, show that the H -sector TFP grows at the slowest rate, followed by the L -sector.

4.3 Internal Calibration

There are 16 parameters left to calibrate. After normalizing θ_L to $1 - \theta_G - \theta_H$ and λ_L to 1, a total of 14 free parameters remain. These make up the 14-dimensional parameter vector

Table 1. Externally Calibrated Parameters

Parameter	Description	Value	Source
<i>Panel A. Preferences</i>			
β	Period discount factor	0.904	Barro and Sala-i-Martin (2004)
β_g	Intergenerational discount factor	0.500	Boar and Lashkari (2021)
σ	Elasticity of substitution across sectoral goods	0.100	Buera and Kaboski (2009)
χ	Frisch elasticity	0.500	Boppart, Krusell and Olsson (2024)
<i>Panel B. Technology</i>			
ϵ	Elasticity of substitution between skilled and unskilled labor, L -sector	2.900	Acemoglu and Autor (2011)
ϕ_L	Unskilled labor intensity, L -sector	0.750	Fang and Herrendorf (2021)
$A_G(0)$	Initial TFP, G -sector	1.000	Normalization
$A_H(0)$	Initial TFP, H -sector	1.000	Normalization
$A_L(0)$	Initial TFP, L -sector	1.000	Normalization
g_G	Average TFP growth per period, G -sector	0.053	Elbadri and Vizcaino (2025)
g_H	Average TFP growth per period, H -sector	0.005	Elbadri and Vizcaino (2025)
g_L	Average TFP growth per period, L -sector	0.037	Elbadri and Vizcaino (2025)

Notes: This table reports parameters calibrated outside the model using external evidence. One period corresponds to 5 years. G , H and L denote the Goods, High-Skill Services and Low-Skill Services sectors.

$$\Theta \equiv \{\theta_G, \theta_H, \lambda_G, \lambda_H, q, \zeta_1, \zeta_2, \zeta_3, \omega, \nu, \eta, \kappa, \eta_c, \kappa_c\}$$

The vector Θ is jointly identified to match patterns of structural change, time use and the skill distribution in postwar United States. The candidate vector that is chosen is the one that yields the minimum discrepancy between model-simulated and empirical data moments. This is accomplished via simulated method of moments (SMM).

At any given time, the model has just one cohort of parents and one cohort of children. Reconciling cross-sectional data with model therefore requires building and tracking synthetic cohorts over time. I operationalize this by partitioning the sample period (1942-present) into six synthetic cohorts. This means that each calendar year is mapped to a unique combination of parent generation and childhood stage of development, as per the timing assumptions of the model. Each parent generation-child stage period lasts for a five-year window. More details on the timing convention can be found in Appendix C.2.

The list of targeted moments includes: (i) the share of skilled workers in the first, third and sixth generations, (ii) the log 90-10 wage per hour ratio of skilled workers in the first and sixth generations, (iii) the first and sixth generation High-Skill Services and Goods

expenditure shares, (iv) the third and sixth generation parental childcare time allocation for parents with children in the first stage of childhood development (ages 0–4), (v) the fourth and sixth generation parental childcare time allocation for parents with children in the second stage of childhood development (5-9 years of age) and (vi) the third generation parental childcare time allocation for parents with children in the third stage of childhood development (10-14 years of age). Appendix C.3 specifies the data sources and details used to construct the 14 targeted data moments.

The choice of data moments is made such that the model reproduces patterns of structural change, time allocation and the skill distribution of the postwar US economy while simultaneously ensuring that the Jacobian matrix is of full rank. Heuristically, one can see this by splitting the parameters and moments into three blocks. The first block is concerned with matching the evolution of structural change in the US. The parameters that belong in this block are $\{\theta_G, \theta_H, \lambda_G, \lambda_H\}$. The consumption technology weights $\{\theta_G, \theta_H\}$ can be mapped to the initial aggregate expenditure shares while the parameters $\{\lambda_G, \lambda_H\}$ are particularly helpful in shaping final expenditure shares given initial shares by virtue of the fact that these two parameters control the income elasticity of demand.

The second block is concerned with matching time use patterns. The parameters that belong in this block are $\{\zeta_1, \zeta_2, \zeta_3, \omega, \nu\}$. The elasticities $\{\zeta_1, \zeta_2, \zeta_3\}$ control the importance of parental childcare time allocation in the human capital law of motion in each stage of childhood development which in turn shapes the continuation value that parents face. Cross-stage variation in childcare time allocation obtained from stage-parental childcare time allocation profiles can pin down stage-specific elasticities. Simultaneously, $\{\omega, \nu\}$ shape the weight and curvature of parental time allocation to childcare in flow utility, thereby determining the direct enjoyment that parents derive from time spent with children. These can be pinned down by the level and trend of childcare time allocation over generations.

The final block targets the supply of skills in the US. The parameters $\{\eta, \kappa, \eta_c, \kappa_c, q\}$ are placed in this block. The initial parent human capital distribution follows a lognormal distribution $G(\eta, \kappa)$ where η and κ are the location and shape parameters, respectively. Conditional on the occupational choice threshold (which is endogenous), η and κ control the share of skilled workers and the variance of earnings conditional on being skilled (proxied by the log 90-10 wage gap conditional on being a skilled worker) in the first generation, respectively. Likewise, fixing the human capital production function parameters and the occupational choice threshold, the lognormal initial child human capital draw distribution $G(\eta_c, \kappa_c)$ can be disciplined by targeting the same moments of the sixth generation (current) US skill distribution. Finally, fixing the parental time investment elasticities in the human capital production function, the human capital persistence parameter q influences the extent to which a given level of parental time investments builds a comparative advantage

in skilled work. It can therefore be disciplined by the skilled workers' share in between the initial and current US skill distributions.

Table 2 presents the values of the internally calibrated parameters. A worker disutility constant of 1.374 suggests that work is relatively distasteful for parents. The CRRA parameter value of 1.11 implies that parental time allocation is indeed treated as a luxury, with greater gross income generating a more than proportional increase in parental childcare time allocation. This value is however slightly lower than those obtained from similar papers that estimate this parameter in the context of time use such as [Bick et al. \(2022\)](#) and [Boppart, Krusell and Olsson \(2024\)](#), but this is expected given that parental childcare time allocation is only one form of leisure. The SMM produces a higher G -sector consumption aggregator weight relative to the H - and L -sectors in order to match the higher expenditure share in this sector in initial periods. Looking at the income elasticity parameters and, coupled with the fact that sectoral goods are complements ($\sigma < 1$), it is clear that the H -sector is the income-elastic sector while the G -sector is the income-inelastic sector. Taken together, the preference parameter values suggest that as TFP rises, the two layers of income effects are simultaneously operating: households are biasing their consumption to H -sector while also increasing the parental childcare time allocation at the expense of work.

Table 2. Internally Calibrated Parameters

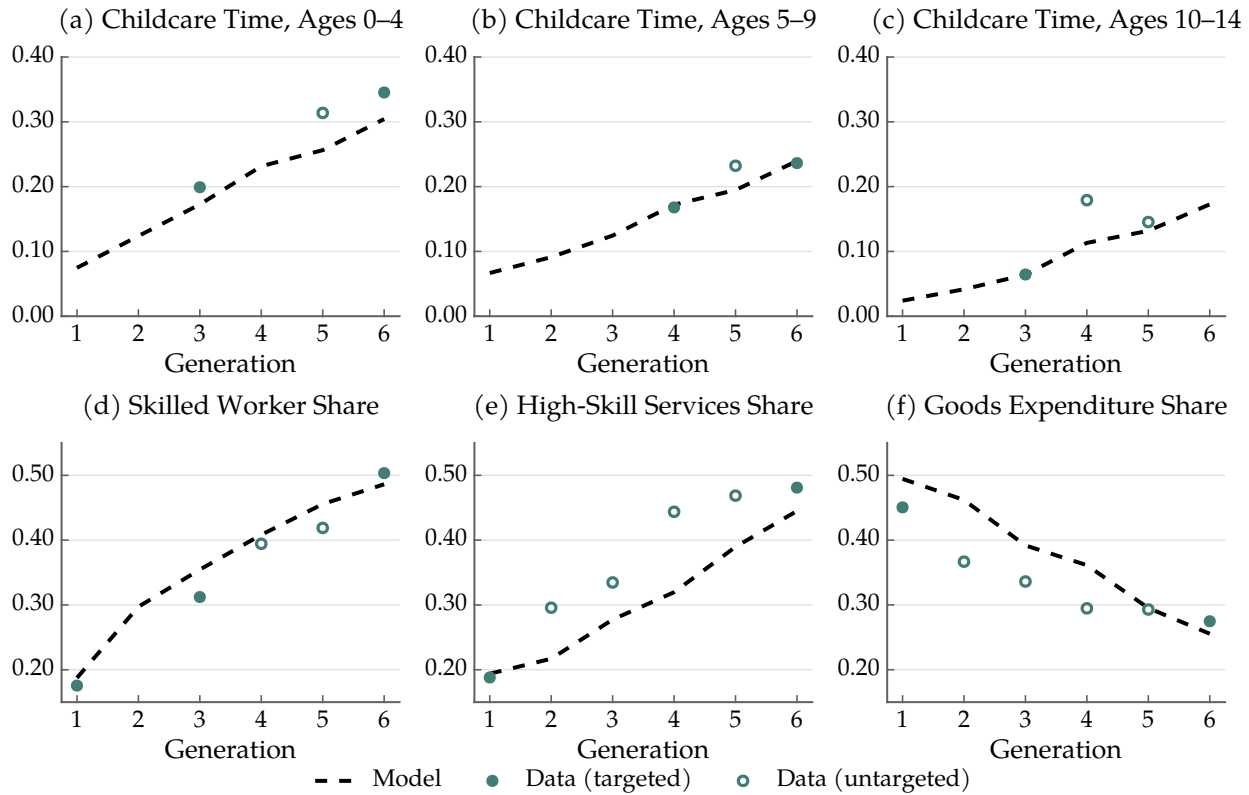
Parameter	Description	Value	Targeted moment
<i>Panel A. Preferences</i>			
ω	Work disutility constant	1.374	Childcare time, stage 1 (Gen. 6)
ν	CRRA parameter	1.110	Childcare time, stage 2 (Gen. 6)
θ_G	Consumption weight, <i>G</i> -sector	0.575	<i>G</i> -sector expenditure share (Gen. 1)
θ_H	Consumption weight, <i>H</i> -sector	0.256	<i>H</i> -sector expenditure share (Gen. 1)
λ_G	Income elasticity parameter, <i>G</i> -sector	0.170	<i>G</i> -sector expenditure share (Gen. 6)
λ_H	Income elasticity parameter, <i>H</i> -sector	1.509	<i>H</i> -sector expenditure share (Gen. 6)
<i>Panel B. Human Capital Technology</i>			
$1 - \zeta_1$	Parental time intensity, stage 1	0.817	Childcare time, stage 1 (Gen. 3)
$1 - \zeta_2$	Parental time intensity, stage 2	0.296	Childcare time, stage 2 (Gen. 4)
$1 - \zeta_3$	Parental time intensity, stage 3	0.001	Childcare time, stage 3 (Gen. 3)
q	Human capital persistence	0.010	Skilled worker share (Gen. 3)
<i>Panel C. Skill Distribution</i>			
η	Location, initial parental distribution	0.321	Skilled worker share (Gen. 1)
κ	Shape, initial parental distribution	1.623	90–10 log wage gap, skilled workers (Gen. 1)
η_c	Location, child endowment distribution	0.399	Skilled worker share (Gen. 6)
κ_c	Shape, child endowment distribution	1.000	90–10 log wage gap, skilled workers (Gen. 6)
<i>Panel D. Normalizations</i>			
θ_L	Consumption weight, <i>L</i> -sector	0.169	Set to $1 - \theta_G - \theta_H$
λ_L	Income elasticity parameter, <i>L</i> -sector	1.000	Reference for λ_G, λ_H

Notes: The 14 parameters in Panels A–C are jointly estimated by simulated method of moments (SMM); the final column reports the data moment most informative for each parameter. The two parameters in Panel D are normalizations. Gen. g denotes generation g ; moments for the skilled worker share, expenditure shares and wage gaps are averages within each generation. *G*, *H* and *L* denote the Goods, High-Skill Services and Low-Skill Services sectors.

Proceeding to the parameters that shape the human capital law of motion, it is clear that across stages of childhood development the importance of parental time investment decreases. In stage 1, the parental time intensity is 0.817 implying that this is the key driver

of human capital accumulation for children when they are young. This number quickly falls in the second and third stages of childhood development to 0.296 and 0.001. This suggests a strong incentive for parents to front-load time investments in their children when they are young, especially since later stages become increasingly reliant on the human capital that is achieved then. This is in part identified by cross-stage variation in parental time investments. The falling parental time intensity resonates with much of the literature that stresses the importance of parental time allocation for early childhood human capital (Del Boca, Flinn and Wiswall, 2014; Francesconi and Heckman, 2016; Yum, 2023; Del Bono et al., 2016). The fact that the human capital persistence reaches a lower-bound corner implies that all the variance in skilled worker wages observed in the data is inherited from the initial child distribution rather than persistence in the human capital production function. The skill distribution parameters provide the location and shape parameters that are required to match the skill distribution in the initial and end generations.

Figure 6. Data vs. Model-Simulated Moments



Notes: Model-simulated moments from the SMM estimation compared with US data moments. Childcare time is the share of parental time (childcare plus paid work) devoted to childcare, for parents whose youngest child is in the corresponding stage of childhood: ages 0–4, 5–9 and 10–14 in panels (a)–(c). Panel (d) reports the share of employment in skilled occupations. Panels (e) and (f) report the High-Skill Services and Goods expenditure shares. Filled circles are data moments targeted in the SMM estimation; hollow circles are untargeted.

Figure 6 shows the extent to which the model with the given parameter selection can replicate patterns of structural change, time allocation and the skill distribution in postwar United States. In all six panels, the x-axis is the generation. As per the timing convention in Appendix C.2, Generation 1 begins from 1942 and lasts until 1956 while Generation 6 lasts from 2017 to 2031. Solid red circles are targeted data moments, hollow red circles are untargeted data moments and the green lines are the model-simulated moments. Starting with parental time allocation, the model is able to replicate the cross-stage variation in both levels and changes in parental childcare time allocation (as a share of childcare and work time) over time as shown in the top row of Figure 6. This is particularly helpful for disciplining the stage-specific parental time elasticities in the human capital production function. Looking at the skill distribution, the model reproduces the skilled worker shares in the initial and final generations. The generation 3 skilled workers share is slightly overstated in the model, explaining the lower-bound corner in the human capital persistence parameter q obtained by the SMM. Finally, the model is also able to replicate structural change in household expenditure away from the G -sector and into the H -sector over time.

5. Quantitative Exercises

This section performs four quantitative exercises using the calibrated version of the model. The first exercise quantifies the role of initial conditions — TFP in particular — in shaping structural change. The second and third exercises focus on quantifying the effect of early childhood development on structural change. Specifically, the second exercise assesses the role of the rise in parental time allocation observed in the US while the third exercise assesses the role of dynamic complementarity in shaping human capital outcomes and, by extension, structural change. The final exercise quantifies the role of human capital inequality in driving structural change.

5.1 Exercise 1 — Initial TFP Level and Structural Change

Motivation. The model’s initial conditions are defined by the initial parent human capital distribution and initial TFP levels (which are equalized across sectors). The latter is effectively the lever that controls the initial level of development of the economy and, together with the sectoral TFP growth rates, is the source of income effects that shape both supply- and demand-side drivers of structural change. To highlight these mechanics, this counterfactual exercise explores how initial TFP influences structural change.

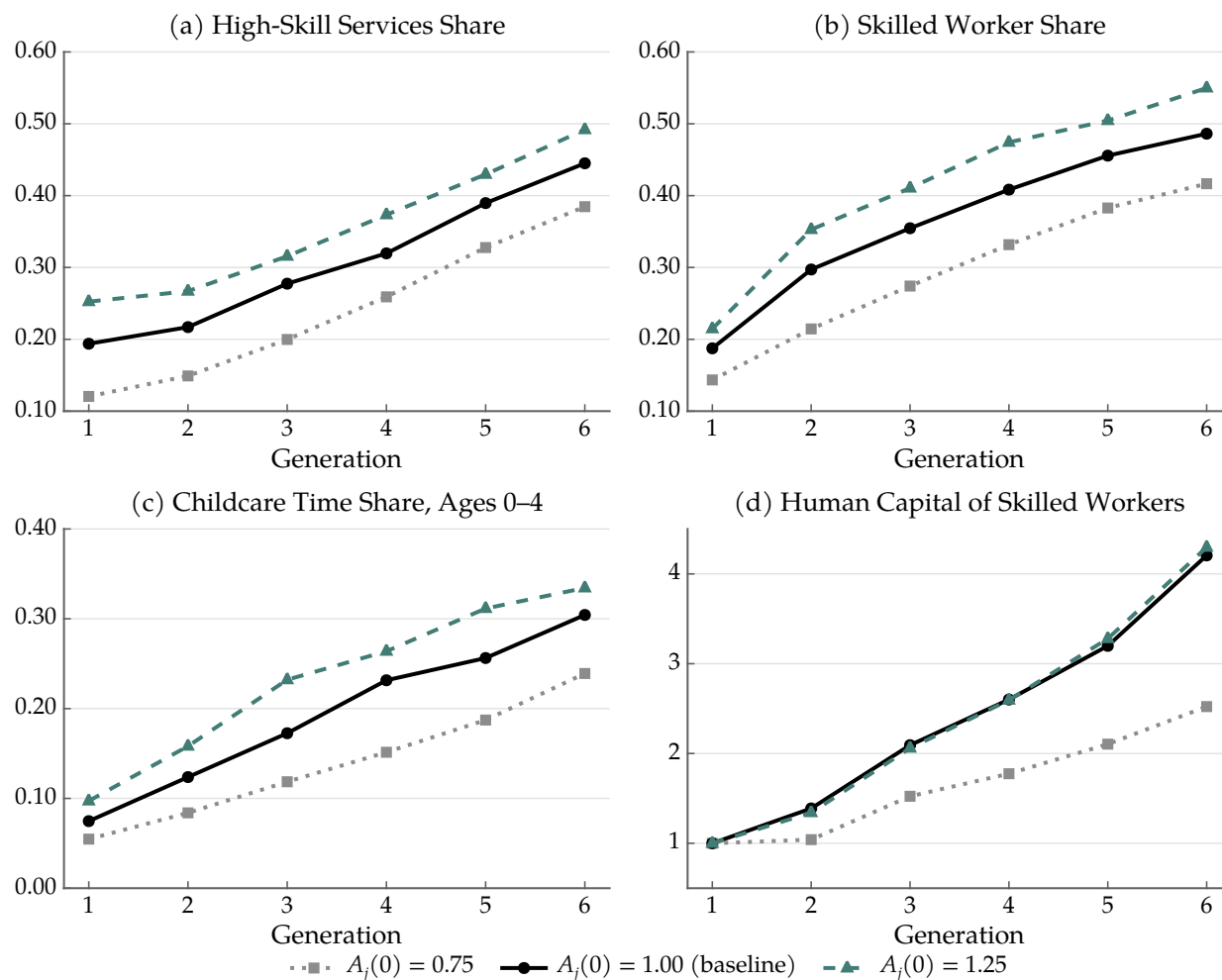
Implementation. Besides the baseline simulation, two counterfactual simulations are run. The first equalizes initial TFP levels across sectors j at 0.75 (i.e. $A_j(0) = 0.75$) while the second equalizes them at 1.25 (i.e. $A_j(0) = 1.25$) compared to their baseline values of $A_j(0) = 1.00 \forall j \in \{G, L, H\}$. For context, the 25 percentage point gap in TFP is broadly in line with the US-Zimbabwe and US-South Africa TFP gap in the 1960s according to the Penn World Tables (Feenstra, Inklaar and Timmer, 2015). All simulations keep the initial parent human capital distribution and TFP growth rates fixed at their baseline xwvalues.

Results. The results of the first exercise are presented in Figure 7. The key takeaway is that initial TFP levels govern the speed of the transitional dynamics. The top-left panel plots the generation-average H -sector expenditure share and shows that for any given generation, a higher initial TFP level raises the H -sector expenditure share. For generation 1, a 25 percentage point increase (decrease) in initial TFP level relative to baseline increases (decreases) the H -sector expenditure by 5.9 (7.4) percentage points relative to the baseline. These level differences are largely persistent; by generation 6, the counterfactual scenario where initial TFP level was 25 percentage points higher (lower) relative to baseline sees a higher (lower) H -sector expenditure by 4.7 (6.1) percentage points relative to baseline. Another representation of this outcome is recorded in the middle-left panel which plots the number of generations required for the generation-average H -sector expenditure share to exceed 40% against the initial TFP level. The higher the initial TFP level, the fewer the number of generations that are required for the H -sector expenditure share to exceed the 40% threshold. These results square with the positive link between the High-Skill Services employment share and the level of development shown in Figure 5.

The top-right panel of Figure 7 shows that together with the rise in the High-Skill Services expenditure share, higher initial TFP levels induce a rise in the share of skilled workers in the economy. Two mechanisms are at play. The first is the rise in the demand for skills. Fixing the parent human capital distribution, higher TFP levels across sectors enrich households. By doing so, households bias their consumption towards the H -sector given that this is the income-elastic sector. With the H -sector being the skill-intensive sector, this raises the demand for skills in the economy. Reinforcing this mechanism is the effect of differential TFP growth rates; slow TFP growth in the H -sector increases the wage of skilled workers in the H -sector, thereby raising the returns to skills. With perfect foresight, parents internalize these dynamics and invest more time in their children given their altruistic nature. The second mechanism highlights the role of income effects in driving greater time investments in their own right. Non-homothetic preferences in the consumption-time allocation margin implies that as households get richer, they invest more time in their children because they enjoy doing so. With greater investments, next-generation

parents enjoy greater human capital, effectively making them richer not only because of counterfactually higher TFP levels, but also because of higher human capital levels. Thus, parents effectively pass on these two dynamics to further generations, consequently driving persistent differences in structural change as a result of differences in initial conditions.

Figure 7. The Role of Initial TFP Levels



Notes: Simulations with initial sectoral TFP, $A_j(0)$, set to 0.75, 1.00 (baseline) and 1.25 in all sectors. All outcomes are averages within each generation. Panel (c) reports the share of parental time (childcare plus paid work) devoted to childcare for parents with children aged 0–4. Panel (d) reports the mean human capital of skilled workers, normalized to 1 in generation 1 of each simulation.

These dynamics can be verified by the rise in childcare time allocation by parents towards their children when their children are youngest. The bottom-left panel of Figure 7 shows that higher initial TFP levels drive greater time investments not just by the initial generation, but also subsequent generations. The 50 percentage point gap in initial TFP levels generates just a 4.3 percentage points in time allocation shares in the first generation

but a 9.5 percentage point difference in the sixth generation. The bottom-right panel shows that these investments materialize into a far more human capital-intensive economy.

5.2 Exercise 2 — Childcare Time Allocation and Structural Change

Motivation. What role does early childhood development play in driving structural change? Motivated by the importance of parental time allocation as an input in the human capital development of young children (Del Boca, Flinn and Wiswall, 2014; Yum, 2023) and the finding in Section 4 that the human capital of children in the first stage of childhood development (i.e. between 0 and 4 years old) is particularly sensitive to parental time inputs, this section quantifies the historical rise in the childcare time allocation of parents towards young children on structural change in the US.

Implementation. This exercise is implemented by fixing the childcare time allocation of all parents with children in the first stage of childhood development (i.e. between 0 and 4 years old) to the generation 1 weighted average value of 7.5% for all subsequent generations. Parents can therefore still make time allocation choices, but only when their children are in the second and third stages of childhood development. These are now conditional on the fixed time input that parents allocate to childcare when the children are youngest. To isolate the effects of the skill distribution, this exercise distinguishes between partial and general equilibrium. The former freezes relative prices to their baseline values while the latter allows relative prices to evolve such that market clearing is achieved.

Results. Table 3 reports the results of this counterfactual exercise. The first column reports the moment of interest. The next two columns report the values from the baseline simulation for the first and sixth generations. The subsequent two columns report the partial equilibrium simulation results (i.e. fixing relative prices and wages to baseline values). The final two columns report the general equilibrium simulation results (i.e. allowing relative prices to evolve such markets clear). The headline result is that the rise in parental time allocation towards young children (i.e. 0-4 years old) can explain 64.3% of US structural change to *H*-sector in partial equilibrium and 35.0% in general equilibrium.

Starting with the partial equilibrium result, the aggregate *H*-sector expenditure share rises from 21.5% to just 30.5% as opposed to from 19.4% to 44.5% in the baseline case between generations 1 and 6. Panel B reveals that this is driven by the sharp decline in skills. With relative prices fixed, the skilled worker threshold across generations is unchanged. Lower parental time investments in young children mechanically compresses the supply of human capital across generations. With the human capital distribution shifting to the left, fewer workers have a comparative advantage in skilled work. Consequently, the share of

skilled workers falls to 11.9% by generation 6 compared to 48.6% in the baseline simulation implying a considerable decline in the supply of skilled workers in the economy. Clearly however, the decline in the supply of skills does not only happen through a fall in the quantity of skilled workers but also through the quality of skilled workers; among skilled workers, the average human capital level among sixth-generation parents falls by almost 50%. Both the quality and quantity margins therefore have a meaningful role in restricting output in *H*-sector by virtue of the fact that this is the skill-intensive sector.

Table 3. Fixing Parental Time Investment in Early Childhood

	Baseline		PE		GE	
	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
<i>Panel A. Childcare Time Share</i>						
Ages 0–4	0.075	0.304	0.075	0.075	0.075	0.075
Ages 5–9	0.067	0.239	0.058	0.146	0.059	0.206
Ages 10–15	0.024	0.173	0.024	0.137	0.024	0.148
<i>Panel B. Human Capital</i>						
Skilled work threshold (Gen. 1 = 1)	1.000	3.649	1.000	3.649	1.000	1.555
Skilled worker share	0.187	0.486	0.187	0.119	0.177	0.468
Mean human capital, skilled (Gen. 1 = 1)	1.000	4.206	1.000	2.148	1.000	0.992
<i>Panel C. Structural Change</i>						
<i>H</i> -sector expenditure share	0.194	0.445	0.215	0.305	0.205	0.368
<i>Panel D. Change from Gen. 1 to Gen. 6</i>						
Δ <i>H</i> -sector expenditure share	0.251		0.090		0.163	
Share of baseline change	1.000		0.357		0.650	
<i>Panel E. Welfare</i>						
Consumption equivalent variation (%)	0.000	0.000	−5.033	−39.142	−6.425	−56.960

Notes: In the counterfactual, parental childcare time for children aged 0–4 is held fixed at its Gen. 1 baseline average (7.5%) in all generations. PE holds relative prices and wages at their baseline values; GE lets them adjust to clear markets. Childcare time shares are fractions of parental time. Skilled work threshold and mean human capital of skilled workers are normalized to 1 in Gen. 1. Panel D reports the change in the *H*-sector expenditure share between Gens. 1 and 6, and that change as a share of the baseline change. Panel E reports the ex-ante dynastic consumption equivalent variation relative to the baseline.

Indirectly, the sharp decline in the supply of skills in the economy means that demand for *H*-sector output is also compressed. With parents effectively poorer because of the decline in human capital levels, non-homothetic preferences are more muted compared to the baseline simulation. This force outweighs any potential increases in income coming from endogenously allocating more time at work. Demand for output from the *H*-sector (which is the income elastic sector) therefore declines. Relative price effects are unchanged

in the partial equilibrium simulation, so the demand-side effects operate only through non-homothetic preferences.

The aggregate welfare implications are reported in Panel E; the distribution-weighted consumption equivalent variation in partial equilibrium implies that in the first generation, parents and their descendants must take a 5.0% reduction in per-period baseline consumption in order to be indifferent between the baseline and the partial equilibrium counterfactual. The decline is sharper for parents and their descendants in the sixth generation (39.1%) now that their human capital has reacted endogenously to the freeze in parental time allocation of young children.

In the general equilibrium simulation, the aggregate *H*-sector expenditure share rises from 20.5% in generation 1 to 36.8% in generation 6 as opposed to from 19.4% to 44.5% in the baseline scenario, respectively. Accordingly, when relative prices and wages are allowed to clear markets in the counterfactual simulation, the rise in US parental time allocation towards the childcare of young children can explain 35.0% of structural change to the *H*-sector. Evolving relative prices lead to a sharp decline in the threshold of skilled work from 3.65 in the baseline simulation to 1.56 in the counterfactual general equilibrium simulation by generation 6, so while the skill distribution still shifts to the left, more workers can access skilled work relative to the partial equilibrium case because of the lower threshold. Consequently, the skilled workers share falls by only 1.8 percentage points in generation 6 relative to the baseline.

Now, however, with parents that have lower human capital accessing skilled work, this weighs down the average human capital level of skilled workers to the extent that it is largely unchanged across six generations. Thus, despite the recovery of the skilled workers shares in the economy, the aggregate welfare implications are larger in magnitude than in the partial equilibrium counterfactual, with the consumption equivalent variation now -6.4% (-57.0%) for first- (sixth-)generation parents and their descendants. The core insight of this exercise is that US parental time investments towards young children has served primarily as a human capital quality amplifier, operating mainly through the quality margin rather than the quantity margin.

5.3 Exercise 3 — Dynamic Complementarity and Structural Change

Motivation. Incorporating the multi-stage human capital production function (6) into a multi-sector growth model is one of the innovations of this paper. A central feature that the production function captures is the notion of dynamic complementarity; the productivity of an additional unit of investment is increasing in the level of human capital that a child has already acquired at any given time. The cumulative nature of the human capital accumulation process makes a strong case for the importance of early childhood

development in explaining the supply of skills in the economy. The aim of this quantitative exercise is to assess numerically how important these incentives are in shaping investment decisions and how these decisions themselves explain the sectoral make-up of a given country.

Implementation. This exercise is implemented by modifying the human capital production function (6). Now, the human capital production will only feature the human capital draw of a child at birth in all stages of childhood development rather than the current human capital of the child, implying the following setup

$$h_i^{g+1}(t+1) - h_i^{g+1}(t) = h_i^{g+1}(gT)^{\zeta_s+q} \left(h_i^g \tau_i^g(t) \right)^{1-\zeta_s}, \quad (26)$$

such that the cross-partial derivative $\frac{\partial^2 h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t) \partial \tau_i^g(t)} = 0$. Intuitively, (26) suggests that the human capital that a child has acquired in their early childhood no longer shapes parental time investment decisions in subsequent stages of childhood development. As a result, any source of dynamic complementarity here comes from the human capital draw at birth rather than the human capital that the child has accumulated. As with the second counterfactual exercise, this exercise distinguishes between partial and general equilibrium. The former freezes relative prices at baseline values while the latter allows relative prices and wages to clear markets.

Results. The results of this counterfactual exercise are presented in Table 4. The first column reports the moment of interest. The next two columns report the values from baseline simulation for the first and sixth generations. The subsequent two columns report the partial equilibrium simulation results (i.e. fixing relative prices and wages to baseline values). The final two columns report the general equilibrium simulation results (i.e. allowing relative prices to evolve such markets clear). The headline result is that dynamic complementarity stemming from past human capital investments from parents can explain 64.0% of US structural change to H -sector in partial equilibrium and 33.1% in general equilibrium with aggregate welfare implications similar in magnitude to the second counterfactual exercise.

Aside from the headline result, the mechanics also resemble those in the second exercise. In the partial equilibrium case, the aggregate H -sector expenditure share rises from 20.4% to 29.5% compared to the baseline values of 19.4% and 44.5% across the six generations. Again, this is mostly driven by a decline in skills, with both the quality and quantity margins playing a role. The share of skilled workers in the economy falls from 48.6% in the baseline simulation to 4.7% in the counterfactual simulation, while the average skill level

among skilled workers declines by almost a half in the same generation. Now, however, parental time allocated to childcare declines across all stages of childhood development, both in levels and in changes across generations. This pattern suggests that shutting down dynamic complementarity across stages substantially weakens parents' incentives to invest in their children. The decline in the skills supply in the economy naturally reinforces this mechanism since the marginal benefit of time investments depend on the human capital of parents. Simultaneously, income effects that would shape the time allocation problem that parents face are now more muted relative to the baseline benchmark.

Table 4. Dynamic Complementarity and Structural Change

	Baseline		PE		GE	
	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
<i>Panel A. Childcare Time Share</i>						
Ages 0–4	0.075	0.304	0.039	0.079	0.046	0.181
Ages 5–9	0.067	0.239	0.035	0.136	0.040	0.151
Ages 10–15	0.024	0.173	0.024	0.134	0.024	0.153
<i>Panel B. Human Capital</i>						
Skilled work threshold (Gen. 1 = 1)	1.000	3.649	1.000	3.649	1.000	1.090
Skilled worker share	0.187	0.486	0.187	0.047	0.185	0.415
Mean human capital, skilled (Gen. 1 = 1)	1.000	4.206	1.000	2.142	1.000	0.689
<i>Panel C. Structural Change</i>						
<i>H</i> -sector expenditure share	0.194	0.445	0.204	0.295	0.200	0.368
<i>Panel D. Change from Gen. 1 to Gen. 6</i>						
Δ <i>H</i> -sector expenditure share	0.251		0.091		0.168	
Share of baseline change	1.000		0.360		0.669	
Contribution of dynamic complementarity	–		0.640		0.331	
<i>Panel E. Welfare</i>						
Consumption equivalent variation (%)	0.000	0.000	–5.894	–41.421	–11.709	–66.914

Notes: The counterfactual replaces the multi-stage human capital technology in Equation (6) with Equation (26), in which human capital at each stage is produced from the child's initial endowment rather than from human capital accumulated in the previous stage. PE holds relative prices and wages at their baseline values; GE lets them adjust to clear markets. Childcare time shares are fractions of parental time. Skilled work threshold and mean human capital of skilled workers are normalized to 1 in Gen. 1. Panel D reports the change in the *H*-sector expenditure share between Gens. 1 and 6, that change as a share of the baseline change, and the contribution of dynamic complementarity (one minus that share). Panel E reports the ex-ante dynastic consumption equivalent variation relative to the baseline.

In the general equilibrium case, the *H*-sector expenditure share rises from 20.0% to 36.8%. Now, relative prices and wages evolve to clear markets, reshaping the returns to skills

and, consequently, incentives to invest in human capital. The skilled workers share rises to reach 41.5% by the sixth generation as a result of the decline in the skilled workers threshold which depends on relative prices, effectively weakening the role of the quantity margin. Like the second counterfactual, the average human capital of skilled workers declines by more than in the partial equilibrium case since less capable workers can now access skilled work. Fundamentally, this counterfactual exercise underscores the importance of dynamic complementarity across stages of childhood development in shaping the aggregate supply of skills in the economy. An important implication of this is that since parental time investment elasticity are largest — by a considerable margin — when children are youngest, early childhood development through dynamic complementarity has been an important engine that has shaped the sectoral composition of aggregate output in the US.

5.4 Exercise 4 — Human Capital Inequality and Structural Change

Motivation. The distribution of human capital is key for the dynamics of the theory. It maps directly into the demand for different sectoral goods as well as the supply of skills in the economy. Non-linear technologies — including non-homothetic preferences and dynamic complementarity in the human capital accumulation technology — imply that the relevant moments of the distribution go beyond just the simple mean and, as a result, the forces that shape structural change will depend in part on the extent of dispersion of human capital in the economy as well as the mean. The purpose of this quantitative exercise is to understand and quantify the role of human capital inequality in shaping structural change.

Implementation. The two parameters that control the extent of human capital inequality in the economy are the initial parent human capital and initial child human capital distribution shape parameters κ and κ_c , respectively. This quantitative exercise simultaneously scales both, effectively modifying the rate of dispersion

$$\hat{\kappa} = \psi\kappa \quad \text{and} \quad \hat{\kappa}_c = \psi\kappa_c \quad \forall \psi \in \{0.65, 1.00, 1.35\} \quad (27)$$

However, since increasing the rate of dispersion would increase the lognormal average level of human capital, I also change the location parameters η and η_c such that the means are preserved at their baseline levels. Mean preservation requires that the new counterfactual location parameter take on the values

$$\hat{\eta} = \eta + \frac{\kappa^2 - \hat{\kappa}^2}{2} \quad \text{and} \quad \hat{\eta}_c = \eta_c + \frac{\kappa_c^2 - \hat{\kappa}_c^2}{2} \quad (28)$$

This ensures that any changes in the simulation results are attributed solely to the change in the distribution of human capital rather than the average.

Results. The results are presented in Table 5. The first column reports the moment of interest. The next two columns report the generation 1 and generation 6 average values from the low inequality simulation where the parameters κ and κ_c are scaled by 0.65 relative to the baseline. The subsequent two columns report the generation 1 and generation 6 average values from the baseline simulation. The final two columns report the generation 1 and generation 6 average values from the high inequality simulation where the parameters κ and κ_c are scaled by 1.35. The key takeaway is that while raising the scaling factor from 0.65 to 1.35 raises the starting average H -sector expenditure share from 14.6% to 26.1%, it slows down the increase in the H -sector expenditure share from 27.7 to 20.3 percentage points between the first and sixth generation.

Starting with the supply side, Panel A reveals that time allocated to childcare — particularly towards younger children — is falling with inequality. This is driven by two forces. First, complementarities in the human capital production function with parent human capital implies that parents have to compensate for the decline in parent human capital by investing more time in their children. Parents in the thicker upper tail in the high inequality simulation can afford to spend less time with their children while simultaneously ensuring that they can access skilled work when they enter the labor market. Second, a more compressed initial child human capital distribution implies that skilled work is more accessible for the larger share of children in exchange for a relatively smaller opportunity cost for parents. Parents take advantage of this by investing more time in their children given that gross income is convex in human capital.

Panel B shows that the initial skilled workers share falls with inequality, which is a natural result of the fact that a more compressed parent human capital distribution is more likely to lead to a higher share of parents exceeding the threshold. By generation 6, the skilled workers share also falls with inequality. This coincides with the fact that more compressed distributions lead to a broad-based increase in parental time allocation towards childcare by parents with young children. This, together with the fact that the skilled work threshold grows fastest in the low inequality case, explains the dramatic rise in the average skill level among skilled workers in the low inequality case relative to the high inequality case and the consequent increase in the distribution-weighted consumption equivalent variation. Thus, high inequality dampens incentives for human capital quality gains, contributing to the slowdown in the rate of structural transformation.

On the demand side, the increasing initial H -sector expenditure share in inequality is a direct artifact of a thicker right tail in the initial parent human capital distribution and

Table 5. Human Capital Inequality and Structural Change

	Low inequality ($\psi = 0.65$)		Baseline ($\psi = 1.00$)		High inequality ($\psi = 1.35$)	
	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
<i>Panel A. Childcare Time Share</i>						
Ages 0–4	0.089	0.312	0.075	0.304	0.057	0.285
Ages 5–9	0.073	0.239	0.067	0.239	0.057	0.233
Ages 10–15	0.022	0.171	0.024	0.173	0.024	0.171
<i>Panel B. Human Capital</i>						
Skilled work threshold (Gen. 1 = 1)	1.000	3.837	1.000	3.649	1.000	3.425
Skilled worker share	0.224	0.494	0.187	0.486	0.145	0.473
Mean human capital, skilled (Gen. 1 = 1)	1.000	5.593	1.000	4.206	1.000	1.924
<i>Panel C. Structural Change</i>						
<i>H</i> -sector expenditure share	0.146	0.424	0.194	0.445	0.262	0.465
<i>Panel D. Change from Gen. 1 to Gen. 6</i>						
Δ <i>H</i> -sector expenditure share	0.277		0.251		0.203	
Share of baseline change	1.104		1.000		0.807	
<i>Panel E. Welfare</i>						
Consumption equivalent variation (%)	2.789	10.069	0.000	0.000	−9.559	−16.577

Notes: The counterfactuals scale the shape parameters κ and κ_c of the lognormal initial parental and child human capital distributions by ψ , adjusting the location parameters η and η_c to preserve both means. Childcare time shares are fractions of parental time. Skilled work threshold and mean human capital of skilled workers are normalized to 1 in Gen. 1 of each specification. Panel D reports the change in the *H*-sector expenditure share between Gens. 1 and 6, and that change as a share of the baseline change. Panel E reports the ex-ante dynastic consumption equivalent variation relative to the baseline. Shares are computed from unrounded values.

non-homothetic sectoral preferences; richer parents with higher human capital initially earn enough to front-load demand for *H*-sector output. The decline in the rate of structural change, on the other hand, results from concavity in the Engel curve. Higher mean-preserving inequality in human capital implies a larger share of parents sit on the higher-end of the human capital distribution, where gains in income translate into smaller rises in the *H*-sector expenditure share relative to parents on the lower-end of the distribution. Taken together, these forces explain the declining rate of structural change to the *H*-sector with human capital inequality.

6. Conclusion

This paper proposes a novel theory whereby early childhood development shapes structural change. I motivate my theory with empirical evidence that the High-Skill Services share, parental time allocation to younger children and early childhood human capital outcomes all rise with development. I then build a multi-sector heterogeneous agents model in general equilibrium to rationalize these facts. In the model, parents are altruistic and enjoy spending time with their children. As sectoral TFP rises, non-homothetic sectoral preferences generate an increased demand for the income-elastic sector (High-Skill Services). Simultaneously, income effects and anticipation of increasing return to skills induce parents to invest more time in their children in early stages of their childhood. Through dynamic complementarity, this increases the supply of skilled workers in the economy which benefits High-Skill Services disproportionately given that this sector employs skills more intensively. Endogenizing supply- and demand-side forces means that the model can account for the feedback effects between both forces.

After calibrating the model to match patterns of structural change, time use and the skills distribution over time in the US, I use the quantitative version of the model to perform several counterfactual exercises. The first quantifies the role of initial TFP levels — the main development lever — in shaping structural change. The second exercise measures the effect of the rise in parental time allocation to childcare of young children across generations on US structural change while the third exercise explores the role of dynamic complementarity in shaping time investments and, by extension, structural change. The final exercise measures the role of human capital inequality in driving structural change.

These results generate key policy implications. The first is that in the context of limited industrialization opportunities and the growing importance of Services in both advanced and developing countries, human capital will prove central for development. Second is that policymakers should not treat human capital as a one-step process; human capital accumulation is not only multi-stage, but also a process where human capital accumulation across stages of childhood development is subject to dynamic complements. This means that early childhood human capital investment amplifies incentives for human capital accumulation in later stages. Policies to improve early childhood human capital are likely to generate large gains for structural change and, ultimately, development going forward.

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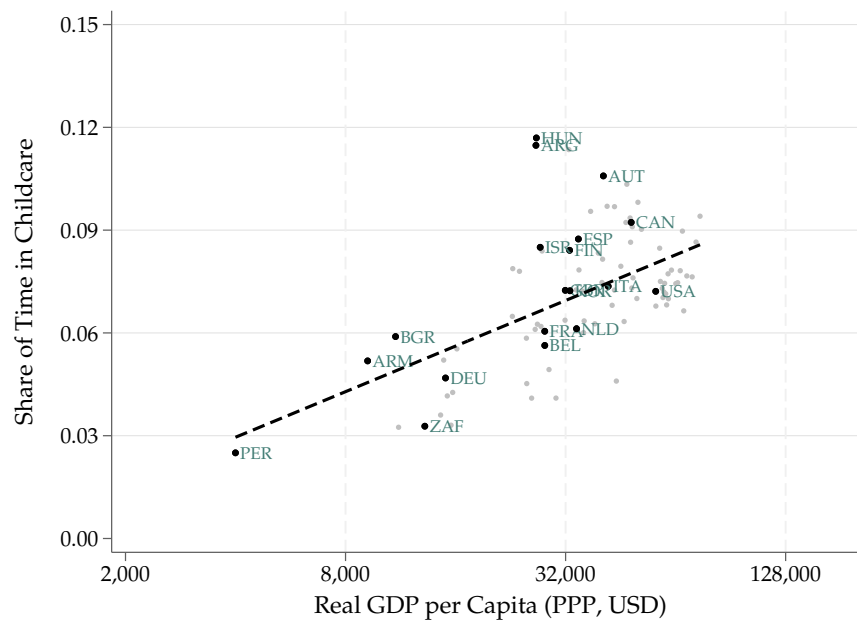
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A. Additional Empirics

A.1 Parental Time Investments in Childcare in Only-Child Households

Figure A.1 restricts attention to diarists who live in a household with a child below the age of 5 and that this is the only child in the household against PPP-adjusted real GDP per capita. The plot reveals that the trend in Figure 1 is robust in one-child households, reflecting a per-child trend.

Figure A.1. Share of Time Spent on Childcare (Only-Child Households)



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

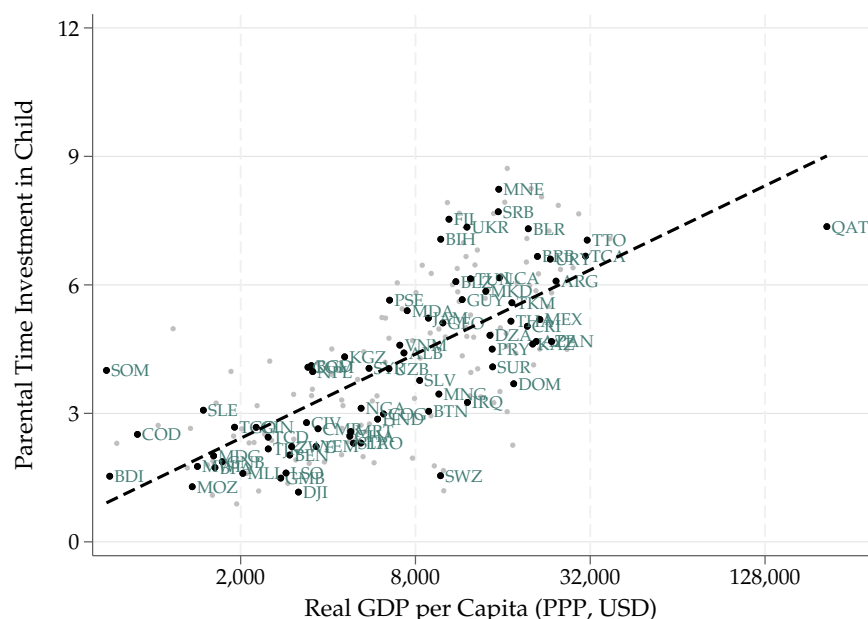
A.2 Parental Time Investments in Childcare Using IPUMS-MICS

The IPUMS-MICS dataset surveys the following six binary questions for mothers and fathers (separately) of children aged between 0 and 4 years old:

- Read books to the child in the last 3 days
- Told stories to child in the last 3 days
- Sang songs to the child in the last 3 days
- Took child outside in the last 3 days
- Played with the child in the last 3 days
- Spent time with child naming, counting, drawing things in the last 3 days

Aggregating the binary answers (1 for yes, 0 for no) for each household and then aggregating at the country-year level using the proposed survey weights yields the parental time investment in child index which is plotted against PPP-adjusted real GDP per capita in Figure A.2. It shows that the cross-country fact that parental time devoted to childcare rises with development even in the lower end of the development spectrum.

Figure A.2. Parental Time Investment in Childcare Index (Using IPUMS-MICS)

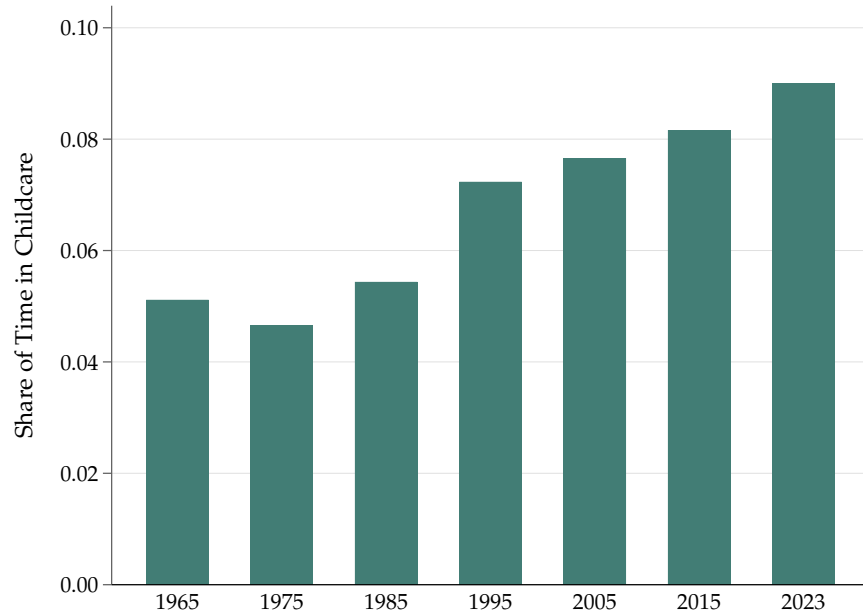


Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.3 Parental Time Investments in Childcare of Younger Children in US

Figure A.3 isolates US time use data from 1965 to 2023 shows that, with the exception of 1975, there is a consistent increase in time spent on childcare over time.

Figure A.3. Share of Time Spent on Childcare Over Time in the US

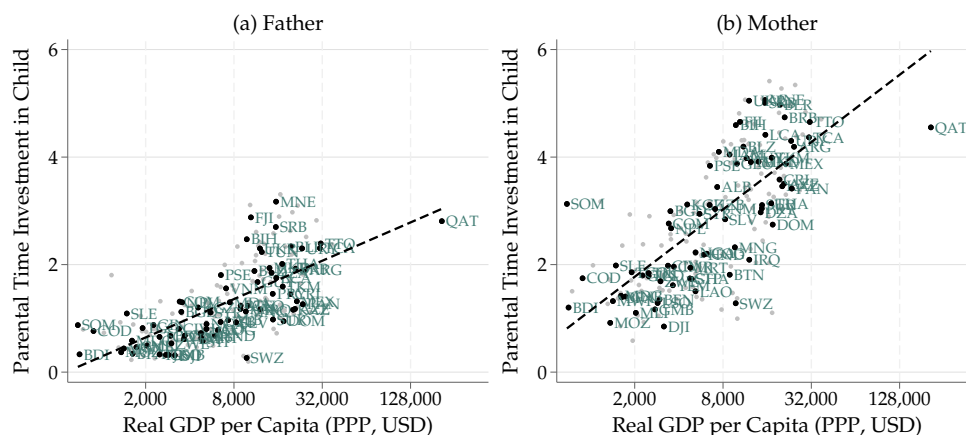


Notes: This chart restricts attention to US samples.

A.4 Parental Time Investments in Childcare of Younger Children By Gender Using IPUMS-MICS

Using IPUMS-MICS data and following the same strategy in Appendix A.2, I plot the parental time investment index by the gender of the parent in Figure A.4. The link shows a very similar trend to Figure 2; both parents spend more time with their children as countries get richer.

Figure A.4. Parental Time Investment in Childcare Index by Gender (Using IPUMS-MICS)

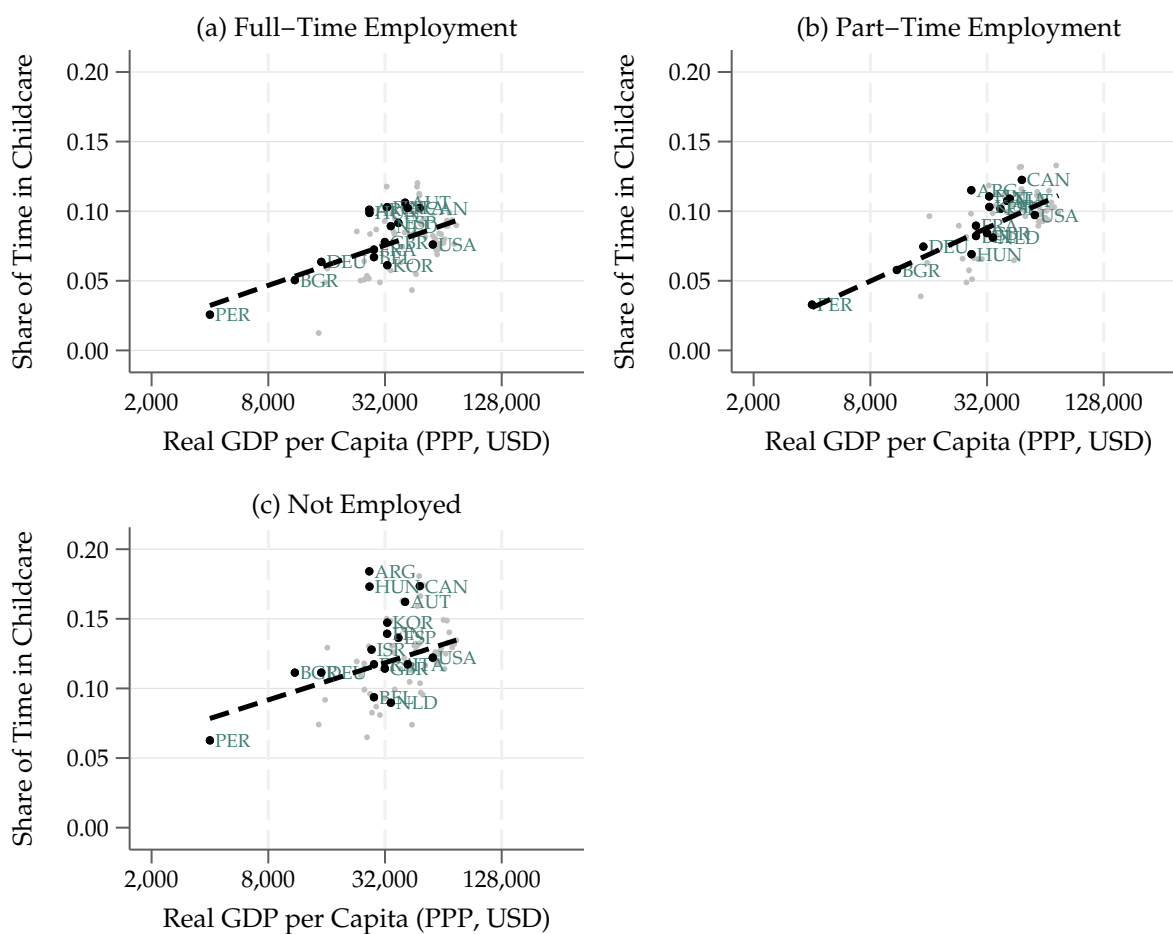


Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.5 Female Time Investments in Childcare of Younger Children By Employment Status

Using time use data, I plot the time allocated to the childcare of young children by females distinguished by employment status against PPP-adjusted real GDP per capita. Figure A.5 shows that the time allocation trend in Figure 2 holds regardless of employment status.

Figure A.5. Female Time Investment in Childcare by Employment Status

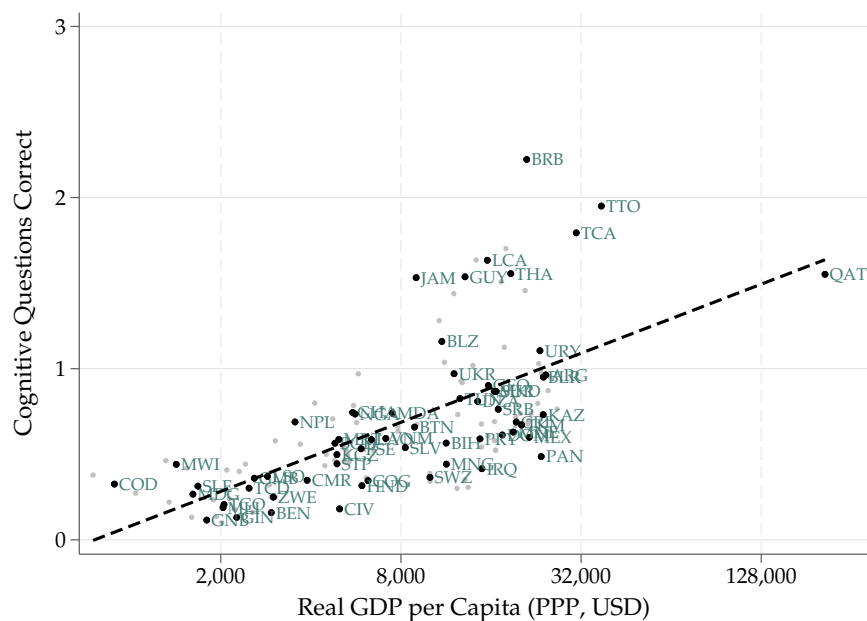


Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.6 Cognitive Skills of 3-Year-Old Children

Figure A.6 plots the average number of cognitive questions that 3-year-old children correctly answer. Similar to Figure 3 that plots the same variable for 4-year-old children, it shows that 3 year-old children in richer countries successfully answer more cognitive questions.

Figure A.6. Early Childhood Cognitive Development Demonstrated (3-Year-Olds)



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.7 Industry Classification

Table A.1 presents the industry classification for Services. The remaining industries make up the Goods sector. This includes Agriculture, Fishing and Forestry, Mining and Extraction, Manufacturing, Construction and Electricity, Gas, Water and Waste Management.

Table A.1. Skill Intensity of Service Industries, United States

Industry	1960	1970	1980	1990	2000	2005	2010	2015	2020	Average
<i>Panel A. High-Skill Services</i>										
Education	0.69	0.64	0.65	0.66	0.69	0.77	0.76	0.77	0.78	0.71
Health and social work	0.52	0.47	0.48	0.54	0.56	0.55	0.53	0.70	0.77	0.57
Financial services and insurance	0.40	0.40	0.45	0.55	0.59	0.69	0.72	0.53	0.59	0.55
Business services and real estate	0.46	0.44	0.49	0.50	0.52	0.54	0.53	0.63	0.68	0.53
<i>Panel B. Low-Skill Services</i>										
Other services	0.32	0.33	0.42	0.47	0.45	0.39	0.33	0.23	0.25	0.35
Public administration and defense	0.15	0.19	0.26	0.29	0.35	0.40	0.37	0.39	0.42	0.31
Wholesale and retail trade	0.20	0.17	0.22	0.26	0.27	0.29	0.29	0.13	0.15	0.22
Transport, storage and communications	0.10	0.10	0.14	0.17	0.18	0.14	0.29	0.28	0.28	0.19
Hotels and restaurants	0.17	0.14	0.15	0.17	0.14	0.20	0.19	0.13	0.14	0.16
Private household services	0.03	0.01	0.03	0.03	0.02	0.01	0.01	0.03	0.21	0.04

Notes: Each cell reports the person-weighted share of employment in an industry (IPUMS INDGEN) accounted for by skilled occupations. Skilled occupations are those classified under OCCISCO as legislators, senior officials and managers; professionals; or technicians and associate professionals. The sample includes employed individuals (EMPSTAT) aged 16–64. The final column is the average over 1960–2020.

B. Model Derivations

B.1 Firm problem

The L -sector producer selects the quantities of $S_L(t)$ and $N_L(t)$ to maximize profits given wages

$$\max_{S_L(t), N_L(t)} \pi_L(t) = p_L(t) A_L(t) \left[\phi_L N_L(t)^{\frac{\epsilon-1}{\epsilon}} + (1 - \phi_L) S_L(t)^{\frac{\epsilon-1}{\epsilon}} \right]^{\frac{\epsilon}{\epsilon-1}} - w_{L,S}(t) S_L(t) - w_{L,N}(t) N_L(t). \quad (\text{B.1.1})$$

The first order conditions imply

$$(1 - \phi_L) p_L(t) A_L(t) \left(\frac{Z_L(t)}{S_L(t)} \right)^{\frac{1}{\epsilon}} = w_{L,S}(t) \quad \text{and} \quad (\text{B.1.2})$$

$$\phi_L p_L(t) A_L(t) \left(\frac{Z_L(t)}{N_L(t)} \right)^{\frac{1}{\epsilon}} = w_{L,N}(t). \quad (\text{B.1.3})$$

Taking the ratio of (B.1.2) and (B.1.3) and cancelling out $Z_L(t)$ gives

$$\frac{N_L(t)}{S_L(t)} = \left(\frac{(1 - \phi_L) w_{L,S}(t)}{\phi_L w_{L,N}(t)} \right)^{\epsilon}. \quad (\text{B.1.4})$$

The G -sector chooses the quantity of $N_G(t)$ to maximize profits given wages

$$\max_{N_G(t)} \pi_G(t) = p_G(t) A_G(t) N_G(t) - w_{G,N}(t) N_G(t). \quad (\text{B.1.5})$$

The first order conditions imply

$$p_G(t) A_G(t) = w_{G,N}(t). \quad (\text{B.1.6})$$

Finally, the H -sector producer chooses the quantity of skilled labor to maximize profits given wages

$$\max_{S_H(t)} \pi_H(t) = p_H(t) A_H(t) S_H(t) - w_{H,S}(t) S_H(t). \quad (\text{B.1.7})$$

The first order conditions imply

$$p_H(t) A_H(t) = w_{H,S}(t). \quad (\text{B.1.8})$$

B.2 Proof for Proposition 1

I drop the generation superscript and dynasty subscript for expositional convenience. Assume a fixed parent human capital level, prices and wages. Let $V^*(y(t))$ denote the value of the parent's problem after maximizing utility conditional on gross income $y(t)$. Furthermore, let $\tau^*(t)$ denote the corresponding optimal time allocation to childcare with a gross income of $y(t)$. The parent's objective function is

$$V^*(y(t)) = \frac{\left(\frac{y(t)(1-\tau^*(t))}{p(t)}\right)^{1-\nu}}{1-\nu} - \omega \frac{(1-\tau^*(t))^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}} + \Phi(\tau^*(t)).$$

where $\Phi(\tau^*(t))$ captures the continuation value consisting of subsequent flow utilities and the present discounted value of future generations' utility under the optimal time allocation $\tau^*(t)$.

Consider now a higher gross income $y'(t) > y(t)$. The optimal time allocation $\tau^*(t)$ remains feasible and the continuation value $\Phi(\tau^*(t))$ is unchanged. Now, however, the parent can benefit from higher utility from consumption since

$$\log\left(\frac{y'(t)(1-\tau^*(t))}{p(t)}\right) - \omega \frac{(1-\tau^*(t))^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}} > \log\left(\frac{y(t)(1-\tau^*(t))}{p(t)}\right) - \omega \frac{(1-\tau^*(t))^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}}$$

It follows that $V^*(y'(t)) > V^*(y(t))$. Accordingly, the parent's intertemporal utility is increasing in gross income.

B.3 Obtaining Marshallian Demands

Given optimal time allocation, each household chooses the consumption bundle that minimizes total expenditure on consumption given the non-homothetic CES aggregator $\forall t \in [gT, (g+1)T)$

$$\min_{\{c_{i,j}^g(t)\}_{t \in [gT, (g+1)T), \forall j \in \{G,L,H\}}} \sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t) \quad \text{subject to} \quad \sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1. \quad (\text{B.3.1})$$

The corresponding Lagrangian is

$$\mathcal{L} = \sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t) + \Lambda(t) \left[1 - \sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} \right], \quad (\text{B.3.2})$$

where $\Lambda_j(t)$ is the Lagrange multiplier. The corresponding first order condition (FOC) is

$$p_j(t) = \Lambda(t) \frac{\sigma-1}{\sigma} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{-1}{\sigma}} \frac{1}{c_i^g(t)^{\lambda_j}}. \quad (\text{B.3.3})$$

Multiplying both sides by the sectoral good $c_{i,j}^g$ implies

$$p_j(t) c_{i,j}^g(t) = \Lambda(t) \frac{\sigma-1}{\sigma} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}}. \quad (\text{B.3.4})$$

To get rid of the Lagrange multiplier, sum (B.3.4) over all j and plug in the non-homothetic CES aggregator

$$\sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t) = \Lambda(t) \frac{\sigma-1}{\sigma}. \quad (\text{B.3.5})$$

Defining $p_i^g(t) c_i^g(t) = \sum_{j \in \{G,L,H\}} p_j(t) c_{i,j}^g(t)$, (B.3.5) can be substituted into (B.3.3) to obtain

$$p_j(t) = p_i^g(t) c_i^g(t) \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{-1}{\sigma}} \frac{1}{c_i^g(t)^{\lambda_j}}. \quad (\text{B.3.6})$$

Rearranging (B.3.6) to solve for the sectoral good yields

$$c_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)c_i^g(t)} \right)^{-\sigma} c_i^g(t)^{(1-\sigma)\lambda_j}. \quad (\text{B.3.7})$$

While (B.3.7) is indeed the Marshallian demand as it expresses sectoral consumption in terms of total expenditure (which is income in equilibrium), prices and parameters, it can be re-expressed as a function of expenditure shares. To see this, multiply and divide both sides of (B.3.7) by the sectoral price $p_j(t)$ and by total expenditure $p_i^g(t)c_i^g(t)$ to obtain

$$\frac{p_j(t)c_{i,j}^g(t)}{p_i^g(t)c_i^g(t)} = \theta_j [p_i^g(t)c_i^g(t)]^{\sigma-1} p_j(t)^{1-\sigma} c_i^g(t)^{(1-\sigma)\lambda_j}. \quad (\text{B.3.8})$$

Since $p_i^g(t)c_i^g(t) = \sum_{j \in \{G,L,H\}} p_j(t)c_{i,j}^g(t)$, it must be that $\sum_{j \in \{G,L,H\}} \frac{p_j(t)c_{i,j}^g(t)}{p_i^g(t)c_i^g(t)} = 1$ which allows (B.3.8) to take on the interpretation of a share. Accordingly,

$$c_{i,j}^g(t) = e_{i,j}^g(t) \frac{p_i^g(t)c_i^g(t)}{p_j(t)} \quad \text{where} \quad e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \quad (\text{B.3.9})$$

To fully characterize (B.3.9), it is necessary to solve for the ideal price index $p_i^g(t)$. To pin down the price index, sum the expenditure share $e_{i,j}^g(t)$ over all j

$$1 = p_i^g(t)^{\sigma-1} \sum_{j \in \{G,L,H\}} \theta_j p_j(t)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \quad (\text{B.3.10})$$

After re-arranging (B.3.10), the ideal price index $p_i^g(t)$ can be written as

$$p_i^g(t) = \left(\sum_{j \in \{G,L,H\}} \theta_j p_j(t)^{1-\sigma} c_i^g(t)^{(\lambda_j-1)(1-\sigma)} \right)^{\frac{1}{1-\sigma}}. \quad (\text{B.3.11})$$

B.4 Proof of Proposition 2

Pre-requisites. The proof requires two ingredients. The first is the sectoral expenditure shares of sector $j \in \{G, L, H\}$ in equilibrium

$$e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \quad (\text{B.4.1})$$

The second is the time derivative of the free labor mobility conditions of skilled and unskilled work. Recall that the free labor mobility conditions, equations (16) and (17), are

$$p_G(t)A_G(t) = \phi_L p_L(t)A_L(t) \left(\frac{Z_L(t)}{N_L(t)} \right)^{\frac{1}{\epsilon}} \quad \text{and} \quad (\text{B.4.2})$$

$$A_H(t) = (1 - \phi_L) p_L(t)A_L(t) \left(\frac{Z_L(t)}{S_L(t)} \right)^{\frac{1}{\epsilon}}, \quad (\text{B.4.3})$$

respectively. Starting with the skilled work free labor mobility condition (B.4.3), taking logs yields

$$\log(A_H(t)) = \log(1 - \phi_L) + \log(p_L(t)) + \log(A_L(t)) + \frac{1}{\epsilon} (\log(Z_L(t)) - \log(S_L(t))). \quad (\text{B.4.4})$$

Then, taking the time derivative of (B.4.4) and rearranging generates

$$\frac{\dot{p}_L(t)}{p_L(t)} = \frac{\dot{A}_H(t)}{A_H(t)} - \frac{\dot{A}_L(t)}{A_L(t)} - \frac{1}{\epsilon} \left(\frac{\dot{Z}_L(t)}{Z_L(t)} - \frac{\dot{S}_L(t)}{S_L(t)} \right). \quad (\text{B.4.5})$$

Repeating for unskilled work free labor mobility (B.4.2) yields

$$\frac{\dot{p}_G(t)}{p_G(t)} = \frac{\dot{p}_L(t)}{p_L(t)} + \frac{\dot{A}_L(t)}{A_L(t)} - \frac{\dot{A}_G(t)}{A_G(t)} + \frac{1}{\epsilon} \left(\frac{\dot{Z}_L(t)}{Z_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right). \quad (\text{B.4.6})$$

Replacing (B.4.5) into (B.4.6) yields

$$\frac{\dot{p}_G(t)}{p_G(t)} = \frac{\dot{A}_H(t)}{A_H(t)} - \frac{\dot{A}_G(t)}{A_G(t)} + \frac{1}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right). \quad (\text{B.4.7})$$

Together, (B.4.5) and (B.4.7) characterize the evolution of relative prices using the two free labor mobility conditions. Notice that these depend on the labor allocations and the TFP growth rates.

Main proof. For the H -sector to dominate, it must be that $R_k(t) \rightarrow 0$ as $t \rightarrow \infty \forall k \in \{L, G\}$ where $R_k(t)$ is the ratio of sectoral expenditure shares

$$R_k(t) = \frac{\theta_k p_k(t)^{1-\sigma} c_i^g(t)^{(\lambda_k-1)(1-\sigma)}}{\theta_H p_H(t)^{1-\sigma} c_i^g(t)^{(\lambda_H-1)(1-\sigma)}}. \quad (\text{B.4.8})$$

Taking logs of $R_k(t)$ yields

$$\log(R_k) = \log\left(\frac{\theta_k}{\theta_H}\right) + (1-\sigma)(\log(p_k(t)) - \log(p_H(t))) + (1-\sigma)(\lambda_k - \lambda_H) \log(c_i^g(t)). \quad (\text{B.4.9})$$

The time derivative becomes

$$\frac{\dot{R}_k(t)}{R_k(t)} = (1-\sigma) \left(\frac{\dot{p}_k(t)}{p_k(t)} - \frac{\dot{p}_H(t)}{p_H(t)} \right) + (\lambda_k - \lambda_H)(1-\sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}. \quad (\text{B.4.10})$$

Since $p_H(t)$ is the numéraire, $\frac{\dot{p}_H(t)}{p_H(t)} = 0$. As a result,

$$\frac{\dot{R}_k(t)}{R_k(t)} = (1-\sigma) \left(\frac{\dot{p}_k(t)}{p_k(t)} \right) + (\lambda_k - \lambda_H)(1-\sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}. \quad (\text{B.4.11})$$

Intuitively, (B.4.11) shows that the ratio of sectoral expenditure shares depends on how relative prices and income effects evolve. With gross complementarities in sectoral consumption (i.e. $\sigma < 1$), a decline in the relative price of sector k is one force in favor of the H -sector dominating as the k -sector would mechanically occupy a smaller share of expenditure. Another force is income effects: $\lambda_k < \lambda_H$ would imply that the H -sector is income elastic relative to the k -sector good, implying that the household biases consumption towards that sector provided that consumption growth is positive (i.e. $\frac{\dot{c}_i^g(t)}{c_i^g(t)} > 0$).

Next, substituting relative price growth rates (B.4.5) and (B.4.7) into (B.4.11) $\forall k \in \{L, G\}$ and recognizing that $g_j = \frac{\dot{A}_j(t)}{A_j(t)}$, I obtain

$$\frac{\dot{R}_L(t)}{R_L(t)} = \underbrace{(1-\sigma)(g_H - g_L) + \frac{(1-\sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{Z}_L(t)}{Z_L(t)} \right)}_{\text{relative price effects}} + \underbrace{(\lambda_L - \lambda_H)(1-\sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}}_{\text{income effects}}, \quad \text{and} \quad (\text{B.4.12})$$

$$\frac{\dot{R}_G(t)}{R_G(t)} = \underbrace{(1 - \sigma)(g_H - g_G) + \frac{(1 - \sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right)}_{\text{relative price effects}} + \underbrace{(\lambda_G - \lambda_H)(1 - \sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}}_{\text{income effects}}. \quad (\text{B.4.13})$$

Equations (B.4.12) and (B.4.13) reveal that the relative price and income effects are influenced by a combination of parameters and labor allocations. The parametric assumptions of this proposition assume that $0 < g_H < g_L, g_G$ and that $0 < \sigma < 1$. That is, TFP growth is slowest in H -sector, a force inducing relative prices in L - and G -sectors to decline over time since all three sectoral goods are complements (i.e. $0 < \sigma < 1$). Relative prices are also influenced by the evolution of labor allocations, although this effect is modulated by the elasticity of substitution between skilled and unskilled work in L -sector production, ϵ . Intuitively, ϵ controls the effect of different skill types in L -sector production on relative prices. With $\epsilon = \infty$, skilled and unskilled labor are perfect substitutes and so the skill mix in L -sector production does not affect the marginal product of labor and, by extension, relative prices so relative prices are driven solely by the TFP growth rates and elasticity of substitution in consumption between different sectors.

Income effects, on the other hand, are governed by the income elasticity parameters: the parametric assumption that $\lambda_H > \lambda_G, \lambda_L > 0$ together with the assumption of gross complementarities in consumption imply that H -sector is the income elastic sector. Consequently, households bias their consumption to H -sector and away from L - and G -sectors with income.

The parametric assumptions of this proposition imply that the following holds $\forall t$:

$$-x_G > (1 - \sigma)(g_H - g_G) + (\lambda_G - \lambda_H)(1 - \sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}, \quad \text{and} \quad (\text{B.4.14})$$

$$-x_L > (1 - \sigma)(g_H - g_L) + (\lambda_L - \lambda_H)(1 - \sigma) \frac{\dot{c}_i^g(t)}{c_i^g(t)}. \quad (\text{B.4.15})$$

Equations (B.4.14) and (B.4.15) simply say that the cumulative effect of the parameter restrictions is a net negative force on both $R_L(t)$ and $R_G(t)$, respectively.

Let $\Delta_L(t) = \frac{(1-\sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{Z}_L(t)}{Z_L(t)} \right)$ and $\Delta_G(t) = \frac{(1-\sigma)}{\epsilon} \left(\frac{\dot{S}_L(t)}{S_L(t)} - \frac{\dot{N}_L(t)}{N_L(t)} \right)$. To proceed, I use the fundamental theorem of calculus. Denote $f_k(t) = \log(R_k(t))$ and $\dot{f}_k(t) = \frac{\dot{R}_k(t)}{R_k(t)} \quad \forall k \in \{L, G\}$. Combining (B.4.12)-(B.4.15) with the definition of $\Delta_k(t)$ gives

$$\dot{f}_k(t) < -x_k + \Delta_k(t) \quad \forall k \in \{L, G\}. \quad (\text{B.4.16})$$

A result of the fundamental theorem of calculus is that

$$f_k(t) - f_k(0) = \int_0^t \dot{f}_k(s) ds. \quad (\text{B.4.17})$$

Replacing (B.4.16) into (B.4.17) and integrating it out yields

$$f_k(t) - f_k(0) < -x_k t + \int_0^t \Delta_k(s) ds. \quad (\text{B.4.18})$$

That is, a sufficient condition for $f_k(t) \rightarrow -\infty$ is that the linear term $-x_k t$ dominate $\int_0^t \Delta_k(s) ds$ as $t \rightarrow \infty$. Dividing (B.4.18) by t , this holds whenever

$$\limsup_{t \rightarrow \infty} \frac{1}{t} \int_0^t \Delta_k(s) ds < x_k. \quad (\text{B.4.19})$$

Condition (B.4.19) is the additional requirement in the $\epsilon < \infty$ case. Since $\int_0^t \Delta_k(s) ds = \frac{1-\sigma}{\epsilon} \left[\log \frac{S_L(t)}{S_L(0)} - \log \frac{X_L^k(t)}{X_L^k(0)} \right]$, it would hold whenever the labor allocations remain bounded away from 0 and ∞ , since then $\frac{1}{t} \int_0^t \Delta_k(s) ds \rightarrow 0 < x_k$; but because the allocation growth rates $\frac{\dot{S}_L}{S_L}$, $\frac{\dot{Z}_L(t)}{Z_L(t)}$, and $\frac{\dot{N}_L(t)}{N_L(t)}$ have no closed form, (B.4.19) is verified numerically along the transition path. In the perfect-substitutes case $\epsilon = \infty$, the factor $\frac{1-\sigma}{\epsilon}$ vanishes, so $\Delta_k(t) \equiv 0$ and (B.4.18) reduces to

$$f_k(t) - f_k(0) < -x_k t, \quad (\text{B.4.20})$$

where it is easy to see that since x_k is a constant, as $t \rightarrow \infty$, $f_k(t) \rightarrow -\infty$. With $R_k(t) = \exp(f_k(t))$, this means that $\exp(-\infty) = 0$ which means that the expenditure share of H -sector asymptotically dominates that of the L - and G -sectors.

C. Quantitative Analysis

C.1 Numerical Solution

To solve the model numerically, I adapt the algorithm in Buera and Shin (2017). The algorithm is composed of two parts. The first part computes the steady state value function and distribution based on a stationarized, one-sector version of the where a single sector dominates (as per Section 3.4). The second part computes the transition by iterating over price sequences until convergence using the steady state as the terminal condition.

The state space is 2-dimensional, defined on parent-child human capital pairs. In recognition that the support for initial childhood human capital grids may differ from the parent human capital support, I distinguish between $\mathcal{H}$ and $\mathcal{H}'_0$ where the former is the support for parent human capital and the latter is the support for child human capital obtained from the initial draws. The state space is therefore defined along the $\mathcal{H} \times \mathcal{H}'_0$ dimensions.

To proceed, I discretize the parent human capital support $\mathcal{H}_d = \{h_1, \dots, h_K\}$ and the child human capital draw support $\mathcal{H}'_{0,d} = \{h'_{0,1}, \dots, h'_{0,K}\}$. I also discretize the initial child human capital draw probability density function $g(\eta_c, \kappa_c)$ into a probability mass function $g_d(\eta_c, \kappa_c)$. Finally, I generate sequences of $A_j(t)$ which are constructed by feeding the constant sectoral TFP growth rates g_j into a compound growth function.

Steady State. Algorithm 1 presents the solution method for solving for the steady state in a single-sector economy with only H -sector output being consumed and produced. This is done in three steps. The first step performs value function iteration over the generation's problem for each parent human capital-child draw pairs. This step pins down the optimal policy function for each pair and the terminal condition, which is the value function in steady state V^{ss} . The second step iterates over the parent human capital distribution using the optimal time allocation policy function and the initial child human capital draws. The third step ensures that the market for H -sector output clears. Both consumption and skills supply are aggregated using the parent human capital and initial child human capital distributions as weights.

Algorithm 1 Steady-State Equilibrium

Require: Grids $\mathcal{H}_d, \mathcal{H}'_{0,d}$, child pmf $g_d(\eta_c, \kappa_c)$, choice set $\mathbb{T}^S$, discount factors β, β_g , tolerance ε , and maximum iterations $i_{\max}$.

Ensure: $V^{\text{ss}}, \mu^{\text{ss}}$, and τ^{ss} .

- 1: Stationarize wages: $w(t)/A(t) = 1$ for all $t > \mathcal{T}$.
- 2: Initialize $V^{(1)}$, set $i \leftarrow 1$ and $\Delta_V \leftarrow \infty$.
- 3: **while** $\Delta_V > \varepsilon$ and $i < i_{\max}$ **do**
- 4: **for** $(h, h'_0) \in \mathcal{H}_d \times \mathcal{H}'_{0,d}$ **do**
- 5: Solve

$$\tilde{V}(h, h'_0) = \max_{\tau \in \mathbb{T}^S} \left\{ \sum_{s=1}^S \beta^{s-1} u(c_s, \tau_s) + \beta^T \beta_g \mathcal{I}[V^{(i)}](h'(h, h'_0, \tau)) \right\}, \quad c_s = (1 - \tau_s)h,$$

and store $\tau^{\text{ss}}(h, h'_0)$ and $c^{\text{ss}}(h, h'_0)$.

- 6: **end for**
 - 7: $V^{(i+1)}(h) \leftarrow \sum_{h'_0 \in \mathcal{H}'_0} g_d(h'_0) \tilde{V}(h, h'_0)$.
 - 8: $\Delta_V \leftarrow \max_{h \in \mathcal{H}} |V^{(i+1)}(h) - V^{(i)}(h)|$; $i \leftarrow i + 1$.
 - 9: **end while**
 - 10: $V^{\text{ss}} \leftarrow V^{(i)}$.
 - 11: Construct Π^{ss} from τ^{ss} and iterate $\mu^{(i+1)} = \Pi^{\text{ss}} \mu^{(i)}$ until convergence; set $\mu^{\text{ss}} \leftarrow \mu^{(i)}$.
 - 12: **for** $s = 1, \dots, S$ **do**
 - 13: $c_s^{\text{ss}} \leftarrow \mathcal{L} \sum_{h, h'_0} \mu^{\text{ss}}(h) g_d(h'_0) c_s^{\text{ss}}(h, h'_0)$.
 - 14: $S_s^{\text{ss}} \leftarrow \mathcal{L} \sum_{h, h'_0} \mu^{\text{ss}}(h) g_d(h'_0) h (1 - \tau_s^{\text{ss}}(h, h'_0))$.
 - 15: Check $|c_s^{\text{ss}} - S_s^{\text{ss}}| < \varepsilon$.
 - 16: **end for**
-

Transition. Algorithm 2 presents the solution method applied to solve for the transition. Crucially, this is achieved by iterating over the fixed point sequences of L - and G -sector prices until markets clear and free labor mobility holds.

Algorithm 3 is called in line 7. This algorithm takes as inputs the proposed price paths, solves the household problem and computes the joint distribution and, accordingly, the aggregate quantities $c_j(t), S(t), N(t) \forall t < \mathcal{T}, j \in \{G, H, L\}$. This is done in three steps. The first step takes as input the steady state value function solved in Algorithm 1 and, through backward induction, computes optimal policy for each household human capital pairs. The second step uses the optimal time allocation policy and shoots forward using the Kolmogorov Forward Equation (KFE) to solve for the joint distribution in all generations. The third step uses the joint distribution to compute equilibrium aggregates including

the supply of skills in the economy and sectoral output demand given the proposed price paths.

Algorithm 2 Transition-Path Equilibrium: Price Block

Require: TFP paths $\{A_j(t)\}_{t \in \mathcal{D}}$, $j \in \{G, H, L\}$, terminal value V^{ss} , initial distribution μ_1 , damping parameters γ , ξ_{lo} , ξ_{hi} , tolerances ε_L , ε_G , and iteration limits.

Ensure: Equilibrium paths $\{p_G(t), p_L(t)\}_{t \in \mathcal{D}}$.

- 1: Set $p_H(t) \leftarrow 1$ for all $t \in \mathcal{D}$. ► Numeraire
 - 2: Initialize $p_G^{(1)}$, $i_G \leftarrow 1$, and $\Delta_G \leftarrow \infty$.
 - 3: **while** $\Delta_G > \varepsilon_G$ and $i_G \leq i_{G,\text{max}}$ **do**
 - 4: Compute $\bar{h}(t)$ for all $t \in \mathcal{D}$.
 - 5: Initialize $p_L^{(1)}$, $i_L \leftarrow 1$, and $\Delta_L \leftarrow \infty$.
 - 6: **while** $\Delta_L > \varepsilon_L$ and $i_L \leq i_{L,\text{max}}$ **do**
 - 7: Obtain $c_j(t)$, $S(t)$, and $N(t)$ from Algorithm 3. Back out $N_j(t)$ and $S_j(t)$.
 - 8: Compute implied prices $p_L^{\text{imp}}(t)$.
 - 9: $\Delta_L \leftarrow \max_{t \in \mathcal{D}} |p_L^{(i_L)}(t)/p_L^{\text{imp}}(t) - 1|$.
 - 10: $p_L^{(i_L+1)}(t) \leftarrow p_L^{(i_L)}(t) - \gamma[p_L^{(i_L)}(t) - p_L^{\text{imp}}(t)]$, $t \in \mathcal{D}$.
 - 11: $i_L \leftarrow i_L + 1$.
 - 12: **end while**
 - 13: Compute implied prices $p_G^{\text{imp}}(t)$ and errors $e_G(t) \leftarrow p_G^{(i_G)}(t)/p_G^{\text{imp}}(t) - 1$.
 - 14: $\Delta_G \leftarrow \max_{t \in \mathcal{D}} |e_G(t)|$.
 - 15: Set $\xi(t) \leftarrow \xi_{\text{lo}}$ if $|e_G(t)| < 0.3$ and ξ_{hi} otherwise.
 - 16: $p_G^{(i_G+1)}(t) \leftarrow p_G^{(i_G)}(t) - \xi(t)[p_G^{(i_G)}(t) - p_G^{\text{imp}}(t)]$, $t \in \mathcal{D}$.
 - 17: $i_G \leftarrow i_G + 1$.
 - 18: **end while**
-

Algorithm 3 Transition-Path Equilibrium: Household and Distribution Block

Require: Candidate paths $\{p_G(t), p_L(t), \bar{h}(t)\}_{t \in \mathcal{D}}$, terminal value V^{ss} , initial distribution μ_1 , grids $\mathcal{H}_d, \mathcal{H}'_{0,d}$, child pmf $g_d(\eta_c, \kappa_c)$, choice set $\mathbb{T}^S$, and discount factors β, β_g .

Ensure: $\{c_j(t), S(t), N(t)\}_{t \in \mathcal{D}}, j \in \{G, H, L\}$.

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1: Set  $\bar{V}_G(h) \leftarrow V^{\text{ss}}(h)$ .
2: for  $g = G - 1$  down to 1 do
3:   for  $h \in \mathcal{H}_d$  do
4:     for  $h'_0 \in \mathcal{H}'_{0,d}$  do
5:       for  $\tau \in \mathbb{T}^S$  do
6:         for  $s = 1, \dots, S$  do
7:           Set  $t \leftarrow (g - 1)S + s$ .
8:           Determine  $o^*(h; t)$  and gross income  $w_s(h; t)$ .
9:           Set  $c_s(h, h'_0; g) \leftarrow (1 - \tau_s)w_s(h; t)$ .
10:        end for
11:         $Q_g(h, h'_0, \tau) \leftarrow \sum_{s=1}^S \beta^{s-1} u(c_s, \tau_s) + \beta^T \beta_g \mathcal{I}[\bar{V}_{g+1}](h'(h, h'_0, \tau))$ .
12:      end for
13:       $\tau^*(h, h'_0; g) \leftarrow \arg \max_{\tau \in \mathbb{T}^S} Q_g(h, h'_0, \tau)$ .
14:      Store  $c^*(h, h'_0; g)$  and  $V_g(h, h'_0) \leftarrow \max_{\tau \in \mathbb{T}^S} Q_g(h, h'_0, \tau)$ .
15:    end for
16:     $\bar{V}_g(h) \leftarrow \sum_{h'_0 \in \mathcal{H}'_{0,d}} g_d(h'_0) V_g(h, h'_0)$ .
17:  end for
18: end for
19: Initialize  $\mu_1(h, h'_0) \leftarrow \mu_1(h) g_d(h'_0)$ .
20: for  $g = 1, \dots, G - 2$  do
21:   Propagate  $\mu_g$  through  $h'(h, h'_0, \tau^*(h, h'_0; g))$  to obtain  $\mu_{g+1}(h)$ .
22:   Set  $\mu_{g+1}(h, h'_0) \leftarrow \mu_{g+1}(h) g_d(h'_0)$ .
23: end for
24: for  $t \in \mathcal{D}$  do
25:   Set  $g \leftarrow \lceil t/S \rceil$  and  $s \leftarrow t - (g - 1)S$ .
26:    $S(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) h (1 - \tau_s^*) \mathbf{1}\{h > \bar{h}(t)\}$ .
27:    $N(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) (1 - \tau_s^*) \mathbf{1}\{h \leq \bar{h}(t)\}$ .
28:    $c_j(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) c_{j,s}^*(h, h'_0; g), j \in \{G, H, L\}$ .
29: end for

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C.2 The Timing Convention

Table C.1 connects generations and stages of childhood development that are used to build the synthetic cohorts for the time use data with calendar years. That way, the generation-level moments such as skilled worker and expenditure shares are 15-year averages over the same calendar years that define a generation. This ensures consistency with the stage-specific childcare time moments which partition that window into three 5-year stages of childhood development.

Generation	Stage of development	Calendar years	Age of child (years, inclusive)
1	1	1942–1946	0–4
	2	1947–1951	5–9
	3	1952–1956	10–14
2	1	1957–1961	0–4
	2	1962–1966	5–9
	3	1967–1971	10–14
3	1	1972–1976	0–4
	2	1977–1981	5–9
	3	1982–1986	10–14
4	1	1987–1991	0–4
	2	1992–1996	5–9
	3	1997–2001	10–14
5	1	2002–2006	0–4
	2	2007–2011	5–9
	3	2012–2016	10–14
6	1	2017–2021	0–4
	2	2022–2026	5–9
	3	2027–2031	10–14

Table C.1. Linking Generations, Stages of Childhood Development, and Calendar Years

Notes: This table maps each model generation and stage of childhood development into the corresponding calendar years and age range. Each generation contains three five-year stages of childhood development: stage 1 covers ages 0–4, stage 2 covers ages 5–9, and stage 3 covers ages 10–14. The mapping is used to identify generation- and stage-specific empirical moments and to construct the synthetic cohorts underlying the parental time-allocation moments.

C.3 Constructing Targeted Data Moments

Time Allocation. I obtain the time allocation moments from the Multinational Time Use Survey (Lamote de Grignon Pérez et al., 2024), restricting attention to the US samples which are available from 1965 onward. I keep individuals who are (i) in full-time employment, (ii) between the ages of 21 and 64 (inclusive) and (iii) live in a household that contains a child below the age of 15.²⁰ To remain consistent with the model, the parental childcare time shares are defined as the time allocated to childcare divided by the time allocated to both childcare and paid work.

Since there exists a single cohort of both parents and children at any given time in the model, all children in the economy are in the same stage of childhood development. I therefore proceed by reconstructing time allocation profiles for synthetic cohorts using age-restricted cross sections. Tracking the time allocation of parents of a given cohort over time (using the timing convention Appendix C.2) enables identification of stage-specific elasticities of parental time allocation. Where multiple calendar years are available for a given parent generation-child stage of development period, I take a simple average. Diarist-day observations for each sample are weighted using the proposed survey weights. The five time allocation moments are:

Generation 3, stage 1 (1972-1976)

- Time allocated to childcare for diarists who live in a household with a child between 0-4 years old

Generation 6, stage 1 (2017-2021)

- Time allocated to childcare for diarists who live in a household with a child between 0-4 years old

Generation 4, stage 2 (1992-1996)

- Time allocated to childcare for diarists who live in a household with a child between 5-9 years old

Generation 6, stage 2 (2022-2026)

- Time allocated to childcare for diarists who live in a household with a child between 5-9 years old

Generation 3, stage 3 (1982-1986)

- Time allocated to childcare for diarists who live in a household with a child between 10-14 years old

²⁰The surveys do not capture how the diarist relates to the child.

Skill Distribution. The skill distribution data moments are computed using microdata obtained from IPUMS USA (Ruggles et al., 2025). Attention is restricted to individuals who are (i) employed, (ii) between the ages of 21 and 64 (inclusive), (iii) work at least 30 hours a week and (iv) worked for at least 40 weeks.²¹ Skilled workers are defined as workers whose occupation is either managerial, professional or technical (Acemoglu and Autor, 2011), where the occupation identifier variable is OCC1950. Skills are assumed fixed after childhood, meaning that the skill distribution is a generation-specific distribution. Thus, where multiple sample calendar years are available for a single generation, I take a simple average. All skill distribution moments are weighted using person weights. The five targeted skill distribution moments are:

Generation 1, 1942-1956

- Share of skilled workers
- Log 90-10 wage per hour ratio of skilled workers

Generation 3, 1972-1986

- Share of skilled workers

Generation 6, 2017-present

- Share of skilled workers
- Log 90-10 wage per hour ratio of skilled workers

Structural Change. The production function does not feature intermediate inputs. Thus, to consistently map consumption expenditures to the data, it is necessary to extract the consumption component of total value added at the broad sector level. Herrendorf, Rogerson and Valentinyi (2013) lay out the procedure to do this in their Online Appendix. For the sake of completeness, I provide a brief summary of these steps in Appendix C.4 together with the time series of consumption value added for my three broad sectors of interest. The procedure requires input-output tables and the total requirements matrix which are made available by the BEA for benchmark years (U.S. Bureau of Economic Analysis, 2025). Where data availability permits, I average over the 15-window for a given generation in order to

²¹Hours worked last week (i.e. HRSWORK1) is not available for all samples. Where unavailable, I proxy hours worked last week by usual hours worked per week (i.e. UHRSWORK). Hours worked per year are proxied by the product of HRSWORK1 and WKSWORK1 where available and UHRSWORK and WKSWORK1 otherwise.

remove stage-specific trends and isolate cross-sectional variation in sectoral expenditure shares. The four targeted structural change moments are:

Generation 1, 1942-1956

- High-Skill Services expenditure share
- Goods expenditure share

Generation 6, 2017-present

- High-Skill Services expenditure share
- Goods expenditure share

C.4 Extracting Consumption Value Added

Herrendorf, Rogerson and Valentinyi (2013) offer a complete description of the steps required to extract sectoral consumption value added from input-output tables in the Online Appendix B. Given that I distinguish between High- and Low-Skill Services and aggregate Agriculture and Manufacturing into Goods, I repeat this exercise using the same input-output tables but with the relevant sectoral split. I provide a brief description of the steps below for completeness.

Let m be the number of commodities produced by the economy and n be the number of industries that produce these m commodities. In input-output tables prior to 1972, $m = n$.

1. From the Use table, extract the final consumption expenditure vector $\mathbf{e}_c$. This is an $m \times 1$ vector where each element i denotes the value of final consumption expenditure spent on the corresponding commodity.
2. From the Use table, extract total output and total value added by industry. Compute the value added share of output by taking the ratio of total value added over total output for each industry. This vector $\mathbf{v}$ is an $n \times 1$ vector.
3. Extract the total requirements matrix $\mathbf{R}$. In post-1972 tables, this is an $n \times m$ matrix. Entry ji of the total requirements matrix gives the dollar value of industry j 's production that is required to deliver one dollar of the domestically produced commodity i to final uses.
4. Consumption value added by industry is then

$$\mathbf{c} = \langle \mathbf{v} \rangle \mathbf{R} \mathbf{e}_c$$

where $\langle \mathbf{v} \rangle$ is the diagonal matrix with vector $\mathbf{v}$ on its diagonal. Element i of $\mathbf{c}$ denotes the consumption value added of the corresponding industry.

5. Aggregate industries into sectors by summing the elements of $\mathbf{c}$ corresponding to each sector as per the industry classification in Section 2. Repeat these steps for each year where input-output tables are available.

These steps are performed using the BEA's Historical Benchmark Input-Output Tables for years where data is available (1947, 1958, 1963, 1972, 1977, 1987, 1992) as well as the annual Input-Output Accounts Data releases for later years (1997-2024).